Following to Lead: Dominant Followership in Sequential Price Competition

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Abstract

Dominant firms are usually thought to lead prices, setting a pace that rivals match. I show the opposite in online retail. Using synchronized high-frequency cross-platform pricing data, I find that Amazon, despite being the dominant platform, is usually priced at or below rivals. Its distinctive role appears in how it responds when rivals move first. After a rival price increase, Amazon is more than three times as likely as other major platforms to raise its own price within 48 hours, while those same rival platforms show almost no added tendency to raise price after an Amazon increase. When Amazon responds, it typically matches or comes close to the rival's new price, and rival price increases persist longer when Amazon matches them. I call this strategy dominant followership. A sequential Bertrand model modified to break ties in favor of the dominant firm explains why this reversal can be incentive-compatible: with a majority of demand at price parity, the dominant firm may prefer matching a rival's higher price rather than undercutting it. Prices rise above the simultaneous-move Nash equilibrium only when the demand-advantaged firm is also the follower. Dynamic extensions show that the anticipated match can make rival initiation profitable despite short-run demand losses, while limited rival response opportunities make the higher price persist. In Q-learning simulations, the same mechanism emerges from profit feedback: the rival's algorithm learns to initiate higher prices, while the dominant firm captures most of the gains. The main insight is that modern algorithmic price competition among asymmetric firms should be modeled as a sequential response process: the relevant object is not only the price a firm sets, but also whose price change becomes observable first and who responds.

Keywords: **Algorithmic Pricing, Artificial Intelligence, Sequential Price Competition, Demand Asymmetry, Reinforcement Learning**

JEL Classification: **L13, L41, C63, C73, L81**

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1 Introduction

The rise of algorithmic pricing is changing the nature of price competition. Pricing algorithms allow firms to characterize not only optimal price levels but also optimal responses to rivals' price adjustments. The algorithmic price-response functions are becoming increasingly important as AI-driven pricing tools make price responses more scalable and adaptive.[1] They have already attracted regulatory attention. For example, the U.S. Federal Trade Commission (FTC) has accused Amazon's internal pricing algorithm "Project Nessie" of using competitors' pricing responses to guide the entire market toward collusive higher prices (Federal Trade Commission 2024).[2]

In this paper, I study how algorithmic price responses reshape competition when firms differ in demand advantage and response reliability. I do so by documenting differences in price-response behavior among major e-commerce platforms and exploring their implications for modeling price competition. Two forces are central to this response channel. The first is demand advantage: consumers may favor one platform because of convenience, switching frictions, or other non-price factors. This demand advantage can change how the platform responds when rivals adjust prices. In online retail, Amazon is a natural place to look for this force: it is the dominant platform, allegedly accounting for more than 82 percent of gross merchandise value in 2022 under the online-superstore market definition used in the FTC complaint. The second is response reliability: some platforms respond more systematically to rivals' price changes than others. The concern is most pronounced when these forces coincide: the demand-advantaged platform is also the reliable responder. In such markets, the dominant platform's responses may make rivals' price increases easier to sustain.

Using synchronized high-frequency cross-platform pricing data, I document and analyze how different platforms respond to rivals' price changes. The data are novel in two respects. First, they track identical product variants across major competing platforms rather than sellers within a single marketplace. Second, prices are collected within the same time window at two-hour frequency, which allows me to recover the sequence of moves and responses usually hidden in standard pricing

[1] Among 419 B2B pricing executives and decision-makers, 65 to 85 percent reported that their organizations expected to adopt generative and agentic AI in pricing within one to three years; see McKinsey, "B2B pricing: Navigating the next phase of the AI revolution," April 7, 2026.

[2] A related California enforcement action alleges coordinated price increases in which one retailer raised its price and another subsequently matched the higher price. Unlike the mechanism studied here, which operates through independently chosen price responses and does not require coordination, the California filing alleges vendor-mediated agreements among Amazon and competing retailers. See *People v. Amazon.com, Inc.*, Memorandum in Support of Motion for Preliminary Injunction (Cal. Super. Ct. 2026), pp. 2–3.

data. The analysis panel contains more than one million synchronized price observations. The data reveal a counterintuitive role for Amazon, the dominant platform in my setting. Amazon does not simply lead prices upward. Instead, it is unusually likely to raise its own price after rivals raise theirs, and those moves occur soon after the rival increase rather than gradually over the following days. When Amazon follows, it often matches or remains close to the rival's new price. Rival increases are then more likely to remain in place than when Amazon undercuts or does not answer. I call this phenomenon "dominant followership." It captures the central combination of forces: the demand-advantaged platform often plays a reactive pricing role. The question is why following can be incentive-compatible for a demand-advantaged firm.

The dominant followership documented in the data can be rationalized in a simple Bertrand model with one seemingly minor modification: one firm has a demand advantage at price parity. One way to microfound this assumption is a lexicographic decision rule: consumers choose the lower price, but when prices are equal, they favor the dominant platform. This keeps the model close to homogeneous-products Bertrand competition, while allowing platform advantages to matter when prices are comparable. The key insight is that the dominant firm retains a majority of demand at price parity, so it does not have a strong incentive to undercut the rival's higher price and may instead prefer to match it, monetizing its existing demand. By contrast, because the rival receives only a minority of demand at price parity, it does have a strong incentive to undercut. But this undercutting incentive pushes prices toward marginal cost only when the rival can react quickly. A market friction that limits the rival's ability to react can therefore sustain equilibrium prices above the simultaneous-move Nash equilibrium. The single-period model captures this friction through leader-follower timing. The infinite-horizon model extends it by giving the rival only occasional response opportunities, while the dominant firm can respond reliably.

The model explains why dominant followership can soften competition when firms understand the incentives they face. In real-world online retail, however, these incentives may be difficult to know in advance. Retailers set prices across large product assortments where demand conditions and rival pricing patterns differ across products and evolve over time. In such environments, firms may rely on learning algorithms that update pricing policies from realized profit feedback rather than from a known model of demand and rival behavior. I model this learning process with Q-learning simulations: firms are not told to follow the dominant-followership strategy but must discover it on their own, from realized profit feedback, if it is profitable. The canonical Q-learning literature models algorithmic price competition as a simultaneous-move game between symmetric firms. I depart from this benchmark in two empirically motivated ways: firms differ in demand advantage, and price competition unfolds sequentially, with the dominant firm responding more reliably[3] to rival price changes than the rival does. It is not obvious whether learning will reinforce or undo the dominant followership mechanism. The channel operates only when the two forces coincide: the demand-advantaged firm must also be the reliable responder. It disappears in the simultaneous-move benchmark, which removes the responder role. In the sequential environment, the dominant firm's matching response teaches the rival that initiating a higher price can be profitable.

Yet the firm that learns is not the firm that gains most. One might expect the dominant firm's algorithm to drive the price increase, since its responses sustain the higher prices. I show that this is not the case. To identify whose learning matters, I allow one firm at a time to use Q-learning while the other follows a myopic best-response rule. The result is counterintuitive: prices rise when only the rival learns, but not when only the dominant firm learns. The intuition is that the dominant firm does not need a learning algorithm to find its role in the mechanism: matching a higher rival price can be profitable as a static best response. The rival faces a different problem. Raising price first is costly in the short run unless the rival expects the dominant firm to match later. The rival therefore needs learning to see that a higher price can pay off over time. Without that learning, its immediate incentive is to undercut rather than initiate the increase. This learning mechanism can disproportionately benefit the dominant firm and reinforce dominance: the rival's algorithm creates the price increase, consumers bear the cost, but the dominant firm captures most of the gains.

The main insight of this paper is that modern price-response algorithms should be modeled as sequential response processes. This challenges the tacit assumption that simultaneous-move games provide an adequate description of long-run retail price competition. I show that the key object in price competition is not only the price a firm sets, but also whose price change becomes observable first and who responds. Simultaneous-move models miss this channel because they

[3] I use "reliable responder" to refer to the model counterpart of the empirical follower role: the firm that can respond to rival price changes more consistently than its rival can.

remove the responder role through which demand advantage shapes incentives and learning. They may therefore fail to approximate long-run outcomes when firms differ in both demand advantage and response reliability.

The paper makes three contributions. First, I provide new empirical evidence that online retail competition has an important sequential component. Using original high-frequency cross-platform data, I show that platforms set prices sequentially and that Amazon, despite being the dominant platform, stands out as an effective follower. Second, I introduce dominant followership as a new response-based mechanism through which demand asymmetry can soften competition. In sequential price competition, a dominant firm can shape market outcomes by following rather than leading. This overturns the standard Stackelberg intuition that advantage belongs to the first mover and shifts attention from observed price leadership to anticipated response. Third, I show that dominant followership can emerge endogenously from algorithmic learning, and that this learning separates who drives the price increase from who benefits from it: the rival's algorithm generates the price increase, but most of the gains accrue to the dominant firm.

These findings also change how algorithmic pricing should be evaluated. Dominant followership need not look anticompetitive in isolation. The dominant firm is not necessarily leading prices upward, communicating with rivals, or committing to a common pricing rule. It may simply be responding to a rival's price as an individually optimal response. I therefore do not label the mechanism as implicit collusion. Each firm is choosing the price response that is privately optimal given its own demand position. The concern is that, once these responses become systematic, they can enter rival algorithms' learning environments and raise market-wide prices. This implies that regulators and modelers should pay attention not only to price levels, but also to response patterns, timing, and which firms' algorithms learn from those responses. For pricing managers, the results suggest caution in evaluating the returns to adopting learning algorithms: a firm's learned pricing policy may help sustain higher prices while allocating much of the gain to a dominant rival.

Related Literature. This paper contributes to the empirical literature on algorithmic pricing. Existing empirical work primarily studies competitive pricing behavior or whether the adoption or use of pricing algorithms affects price levels or margins in particular markets, including automobiles, gasoline, housing, online marketplaces, and earlier digital-pricing settings (Sudhir 2001; Kannan and

Kopalle 2001; Assad, Clark, Ershov, and Xu 2024; Calder-Wang and Kim 2023; Musolff 2024; Zhao and Berman 2025). Less is known about how algorithmic pricing operates through responsiveness: whether firms react to rivals' price changes within a given response window, and how these reactions vary with firms' demand advantages. I document these response patterns directly in online retail, focusing on algorithmic responsiveness as a channel through which pricing technology can shape market-wide outcomes.

This paper also contributes to the empirical literature on e-commerce price competition. Existing work documents price dispersion, Buy Box competition, and algorithmic repricing within online marketplaces, especially Amazon Marketplace and online retail (Cavallo 2018; Chen, Mislove, and Wilson 2016; He, Hollenbeck, and Proserpio 2022; Lam 2023; Hanspach, Sapi, and Wieting 2024; Musolff 2024). The Project Nessie allegations point to a different object of study: cross-platform pricing responses, in which one online retailer's price changes may affect pricing decisions by other retailers. Studying this setting is difficult because Amazon's prices are relatively visible, while rival retailers' prices are harder to observe at high frequency. I address this gap by constructing synchronized high-frequency pricing data that track identical products across major online retailers. This allows me to measure how price changes propagate sequentially across retailers.

The paper adds to the theoretical literature on algorithmic pricing. Canonical studies show that independently learning algorithms can generate supracompetitive outcomes in simultaneous-move pricing environments (Calvano, Calzolari, Denicolò, and Pastorello 2020; Banchio and Mantegazza 2023; Asker, Fershtman, and Pakes 2024). These settings are less suited to markets in which posted prices are observable and firms can react to rivals' posted prices at different times, with demand realized between price adjustments. This paper shifts attention from algorithmic collusion in simultaneous or symmetric environments to systematic algorithmic responsiveness under asynchronous price adjustment.

Related work relaxes symmetry on the supply side, showing that a faster algorithmic firm can use persistent response rules to induce higher prices from a slower or myopic rival (Brown and MacKay 2025). My paper studies a different asymmetry: demand advantage at comparable prices. The key difference is that speed matters here only because of what the responder gains by moving second. A demand-advantaged follower can find it more profitable to match a rival's higher price than to undercut it, while a less demand-advantaged rival would have a stronger incentive to undercut. The mechanism is therefore not commitment to punish a rival, but an incentive-compatible response that the rival can learn.

This paper is also related to the literature on price-matching guarantees (Edlin 1997; Corts 1997; Hviid and Shaffer 1999; Chen, Narasimhan, and Zhang 2001). That literature studies explicit policies under which firms commit to match rivals' lower prices. My setting differs because the response is not an advertised guarantee, but an algorithmic pricing pattern observed in market behavior. The central object is also different: I study why a demand-advantaged firm may find it profitable to match rival price increases, and how rival algorithms can learn from that response.

The paper also contributes to work on sequential price competition and price leadership. Maskin and Tirole (1988) provide a canonical model of alternating-move price competition, in which firms condition their pricing decisions on rivals' current prices. Klein (2021) extends this sequential pricing framework to Q-learning agents. A related empirical literature studies price leadership, showing that dominant firms can use leadership and price experiments to create focal points and coordinate market prices (Byrne and de Roos 2019). In contrast to this emphasis on dominant-firm leadership, the empirical patterns in this paper motivate a focus on dominant-firm followership: Amazon frequently responds to rivals' price changes. To capture this pattern in a learning framework, I extend sequential Q-learning models by introducing empirically grounded demand and timing asymmetries. The resulting framework shows that, when the follower has a demand advantage, moving second can be more valuable than leading.

Finally, the paper connects algorithmic pricing to the literature on demand-side frictions and demand advantages. Search frictions, product differentiation, loyalty, and switching costs can soften price competition by making consumers less willing or able to switch across firms (Hotelling 1929; Diamond 1971; Varian 1980; Narasimhan 1988; Beggs and Klemperer 1992; Hollenbeck, Hristakeva, and Uetake 2024). This literature mainly studies how these frictions affect price competition and market power. I show that they can also shape the roles firms play in sequential algorithmic competition. In my setting, demand advantage matters not only because it weakens consumer substitution, but because it changes which pricing responses become profitable and learnable.

2 Sequential Price Responses Across Platforms

This section builds the empirical case for dominant followership. A platform with a demand advantage, meaning consumers favor it for reasons other than price, such as brand trust, membership programs, or convenience, should be able to price above its rivals without losing much business. Amazon is a natural setting to look for this kind of advantage: it is the dominant platform in this market, allegedly accounting for more than 82 percent of gross merchandise value in 2022 under the online-superstore market definition used in the FTC complaint (Federal Trade Commission 2024). Yet Amazon is not typically the high-price platform; if anything, it is usually priced at or below its rivals. This is the empirical puzzle: why would the platform best positioned to leverage a demand advantage instead compete so aggressively on price? I answer this puzzle by turning from Amazon's price level to its price responses: Amazon's demand advantage shows up not in charging more, but in how it reacts when a rival does. I show that Amazon is unusually likely to raise price soon after a rival raises its own, typically by matching the rival's new price rather than undercutting it. This response pattern lets Amazon benefit from higher prices without ever having to set them first.

2.1 High-Frequency Cross-Platform Pricing Data

Studying price responses requires data that capture a part of competition usually hidden in standard pricing data: the order in which prices change across competing platforms. To recover this order, I build a high-frequency scraping system that tracks identical products across the full set of identified platforms selling each matched variant, collecting prices within synchronized time windows.

Three design features are central. First, I match products at the exact variant level, including SKU, model specifications, and color, so observed price differences reflect platform pricing rather than product heterogeneity. Second, prices for a given product are collected across platforms within the same time window, at roughly two-hour frequency. This synchronization makes it possible to compare prices for the same product across platforms at nearly the same moment. In 91.3% of price-change bins, exactly one eligible platform changes price, so the panel can recover an order of moves rather than only comparing prices at a point in time.[4] Third, for Amazon and

[4] I define a price update as a change from the platform's most recent observed price for the same product. To avoid treating long gaps as immediate responses, I use only cases where the prior observation occurs within the preceding one to six hours.

Walmart, where the scrape observes marketplace seller identity, I restrict observations to first-party platform listings. This is important because the paper studies platform pricing, not seller pricing within a marketplace. For the other major platforms, the scraped pages do not provide a separate marketplace seller field, so I treat the observed listed price as the platform price.

The panel covers a broad set of consumer electronics and home-appliance categories, including headphones, wearables, smart-home devices, and vacuum and floor-care products, from November 2024 to June 2025.[5] I begin with prominent Amazon-listed products within these categories and keep only products also sold by at least one non-Amazon platform; because the relevant competitors differ across products, the resulting platform set varies and spans both broad-selection online superstores and narrower specialty or brand-specific retailers. After dropping observations with missing prices or marked out of stock, the resulting platform-pricing panel contains 216 matched products, 792 product-platform pairs, and about 1.02 million in-stock price observations. Products are observed on 3.67 platforms on average, and the average scrape interval is 2.3 hours. The product set covers a broad range of price points and demand levels: median product prices range from below \$50 to above \$500, and Amazon's "Bought in Past Month" demand proxy ranges from 50 to 100,000.

2.2 Empirical Puzzle: Amazon's Price Position

If Amazon's demand advantage translated into pricing power in the usual way, it should rarely be the cheapest option on the page. Figure 1 plots Amazon's price relative to the lowest rival price, using observations in which Amazon and at least one rival list the same product.

[5] Categories are selected based on sufficient price variation to study adjustment behavior.

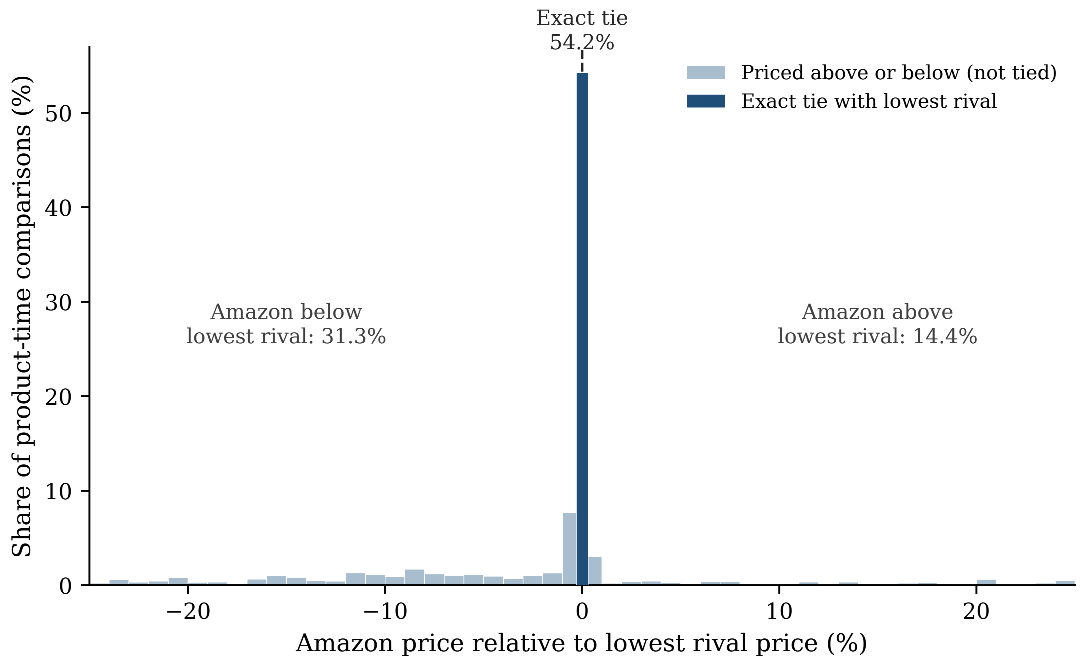

Figure 1. Amazon's Price Relative to the Lowest Rival Price.

Notes: The figure plots Amazon's price relative to the lowest rival price across product-time comparisons in which Amazon and at least one rival have listed prices. A value of zero denotes an exact tie with the lowest rival. Negative values mean Amazon is priced below the lowest rival; positive values mean Amazon is priced above the lowest rival. The figure plots gaps between -25% and 25%; 10.4% of comparisons fall outside this range and are not shown.

Amazon is exactly tied with the lowest rival in 54.2% of comparisons, priced below it in 31.3%, and priced above it in only 14.4%; more than three quarters of comparisons fall within 10% of the lowest rival price. This pattern is stable over time: Amazon's weekly lowest-price share ranges from 77.5% to 88.9% across the sample period, consistently above the weekly mean lowest-price share among major rivals (Walmart, Target, Best Buy, and Home Depot), which ranges from 40.4% to 60.2%.

Amazon's distinctive market role, then, is not expressed through a persistently higher price. It shows up instead in how Amazon responds when rivals move first.

2.3 Amazon Is More Likely to Raise Price After Rival Increases

Amazon is unusually likely to raise price soon after a rival does. Table 1 compares, for each Amazon-product observation, moments when a rival has just raised price against moments when no rival has, using whether Amazon raises price within the next 48 hours as the outcome.[6] Amazon increases within 48 hours in 10.2% of cases when no rival has just raised price, compared with 30.7% of cases when a rival has. The difference is 20.4 percentage points. The same pattern remains after controlling for product, platform, week, and day-of-week differences: Amazon is 15.9 percentage points more likely to raise price after a rival increase ($p < .001$; Panel A of Appendix Table A.1). The broad response pattern therefore shows that rival increases predict later Amazon increases.

Table 1. Raw Increase Probabilities After Rival Price Increases

Platform	P(Increase within 48h | No rival increase)	P(Increase within 48h | Rival increase)	Difference (p.p.)
Amazon	10.2	30.7	20.4
Walmart	8.5	13.5	4.9
Target	3.5	9.7	6.2
Best Buy	7.5	9.6	2.1
Home Depot	5.9	15.2	9.3

Notes: Entries report raw percentages in the main response-regression sample. The first column reports the probability that the focal platform increases its price within 48 hours when no rival increases price at time t. The second column reports the same probability after at least one rival selling the same product increases price at time t.

This broad pattern does not by itself show where the pricing sequence begins. A rival increase may already be a response to an earlier price increase by another platform. I therefore restrict attention to isolated rival increases, in which the ordering is unambiguous: no platform raised price for the product in the prior 48 hours, exactly one identified rival platform (for Amazon, any identified non-Amazon platform selling the product) then raises price, and the responding platform's own price is still unchanged at that moment.[7]

Amazon's response to rival increases is not mirrored by its rivals. Amazon is 12.9 percentage points more likely to raise price after an isolated rival increase, compared with only 3.7 points for the pooled non-Amazon platforms responding to their own isolated rival increases. The gap is 9.3 percentage points (Table 2; $p < .001$). Reversing the direction makes the asymmetry starker: the same non-Amazon platforms show essentially no response to an isolated Amazon price increase, only 0.15 percentage points ($p = .837$). This asymmetry helps address a common-cost-shock explanation.

[6] Among the response horizons I consider, the 48-hour window captures the largest share of later same-product adjustments, 33.9%.

[7] The analogous definition applies to isolated rival decreases.

Table 2. Responses to Isolated Rival Price Increases

	Price increase
Response to isolated rival price increase	
Amazon	12.94
Non-Amazon major platforms	3.65
Difference: Amazon − non-Amazon	9.29
	(1.73), $p < .001$
Observations	597,334
Response to isolated Amazon price increase	
Non-Amazon major platforms	0.15
	(0.73), $p = .837$
Observations	406,877

Notes: Entries are percentage-point effects on increasing price within 48 hours after an isolated price increase, relative to otherwise comparable periods without one. In the top panel, each row defines the isolated rival increase from the perspective of the responding platform. The Amazon row uses isolated non-Amazon increases facing Amazon; the non-Amazon row pools Walmart, Target, Best Buy, and Home Depot when they face isolated increases by their own rivals. The difference row tests equality between these two coefficients, with its standard error, clustered by product, in parentheses. The bottom panel reverses direction and reports the pooled response of the same four platforms following an isolated Amazon price increase. All specifications include product, platform, week, and day-of-week fixed effects; standard errors are clustered by product.

If isolated price increases mainly reflected product-level shocks affecting all platforms selling the same product, response probabilities should not differ so sharply by the identity of the responding platform. Panel B of Appendix Table A.1 shows this pattern holds firm by firm: none of Walmart, Target, Best Buy, or Home Depot approaches Amazon's response individually.

The isolated-event response is also strongest among higher-demand products: it rises from 8.3 percentage points among low-demand products to 10.9 among middle-demand products to 15.5 among high-demand products (Appendix Table A.2). This pattern motivates the model below. When more demand is at stake, responding to a rival's price increase can be more valuable for a demand-advantaged platform. A continuous version of this test, interacting the treatment with log product demand, points in the same direction (1.9 percentage points per log-point of demand, $p = .057$).

Amazon's outsized response to rivals is also robust to the choice of response window. Restricting to isolated rival price increases, Amazon's response remains significantly larger than the pooled response of the other major platforms across alternative response windows: the gap is 4.4 percentage points at 12 hours (p = .001), 5.3 at 24 hours (p = .001), 9.3 at 48 hours ($p < .001$), and 12.0 at 72 hours ($p < .001$). The pattern also holds across product categories, as shown in Appendix Figure A.1.

2.4 Amazon Raises Price Soon After Rival Increases

Amazon is unusually likely to raise price after isolated rival increases. The next question is whether this pattern is tied to the rival's move. If Amazon were already on an upward path, the same events should predict Amazon increases before the rival move. If the estimate only reflected Amazon's frequent repricing, Amazon's increases should be spread throughout the response window rather than concentrated soon after the rival move.

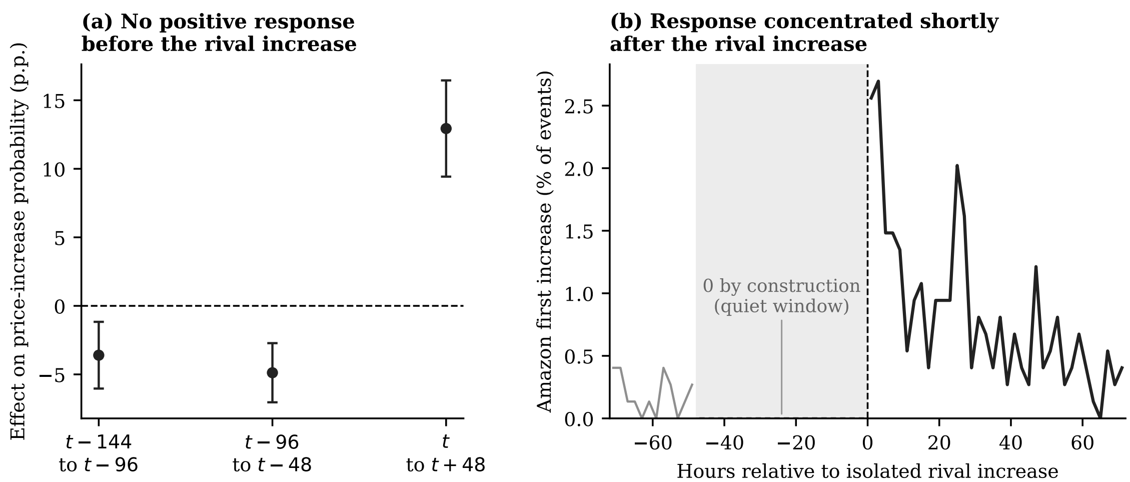

Figure 2. Timing Evidence Around Isolated Rival Price Increases.

Notes: Panel (a) reports Amazon's percentage-point effect on increasing price within 48 hours of an isolated rival increase. Placebo windows assign the same rival-increase events to fake event times 96 or 144 hours earlier. The placebo specifications include product, week, day-of-week, and platform fixed effects, with standard errors clustered by product. Panel (b) conditions on isolated rival increases facing Amazon and plots the share of events for which Amazon's first subsequent price increase occurs in each two-hour bin; the vertical dashed line marks the isolated rival increase. The quiet window is zero by construction because the isolated event requires no same-direction price increase for the product during the prior 48 hours.

Figure 2 addresses both alternatives. Panel (a) shows no comparable positive response in the fake earlier windows. Panel (b) shows that Amazon's first increase is concentrated shortly after the rival move. The timing evidence therefore supports the follower interpretation: Amazon is not simply continuing a prior price increase or repricing often in general; it is raising price after the rival's observed increase.

2.5 Amazon Matches When Responding to Rival Price Increases

Amazon raises price soon after isolated rival increases. The next question is where Amazon's response lands. If Amazon matches the rival's new price, the higher price stands. If Amazon remains below the rival, Amazon's lower price limits the increase. I therefore measure Amazon's response price relative to the rival's new price.

I classify Amazon's first price increase within 48 hours as below, at parity with, or above the rival's new price. For example, if Walmart raises the price of a headphone from $98 to $109 and Amazon later moves from $98 to $109, I classify Amazon's adjustment as a parity response. If Amazon moves to $108, I classify it as below the rival's new price; if it moves to $110, I classify it as above.

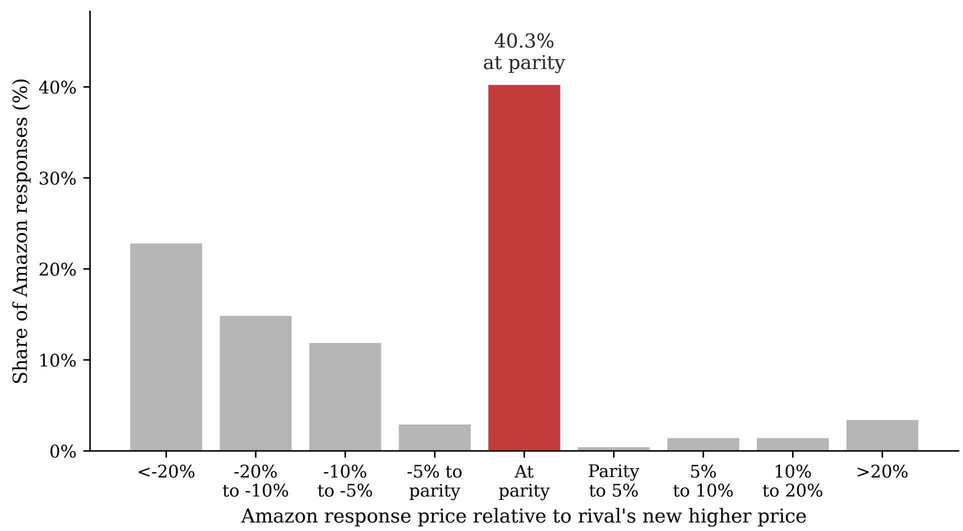

Figure 3. Amazon's Response Price Relative to the Rival's New Price.

Notes: The figure conditions on isolated rival price increases facing Amazon for which Amazon raises price within 48 hours. It plots Amazon's response price relative to the rival's new price. Gaps are measured in percentage points relative to the rival's new price; the parity bar includes responses classified as matches under the tolerance described in the text.

The largest mass in Figure 3 is at parity: 40.3% of Amazon's responses match the rival's new price. Including responses below the rival's new price, 93.0% are at or below the rival's new price. The important point is not price matching by itself, but the matching response of Amazon, the demand-advantaged platform, after a rival moves first. A substantial share of these responses preserve the rival's higher price rather than undoing it through undercutting.

2.6 Rival Increases Last Longer When Amazon Matches

I next ask whether rival increases last longer when Amazon matches them, and whether the same pattern holds for rival decreases. For each isolated rival price change facing Amazon, I compare two cases: Amazon matches the rival's new price within 48 hours, or Amazon does not match it within 48 hours. I then track whether the rival's price remains on the new side of the price change over the following 60 days. If Amazon's matching response helps preserve the rival's higher price, then rival increases followed by an Amazon match should last longer than rival increases not followed by an Amazon match.

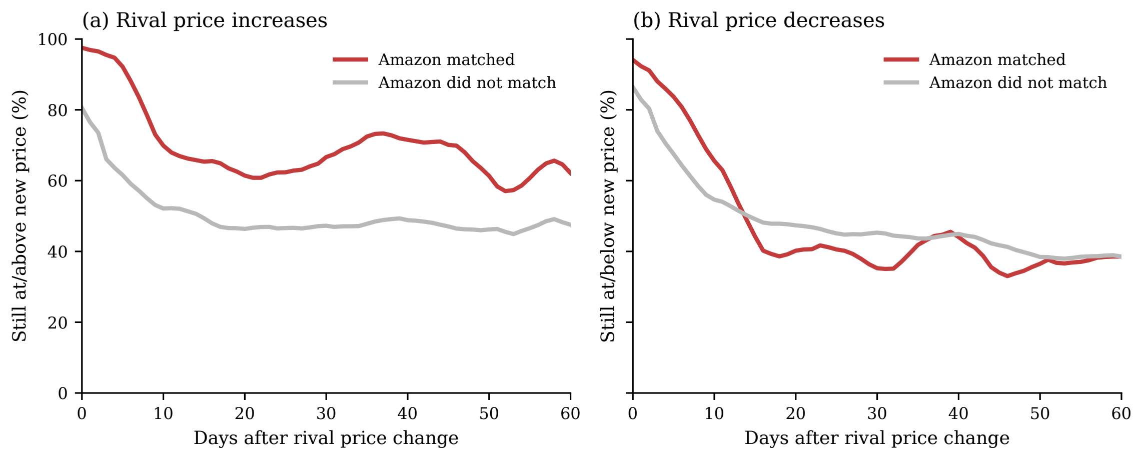

Figure 4. Persistence of Rival Price Changes by Amazon's Response.

Notes: The figure conditions on isolated rival price changes facing Amazon. Panel (a) compares rival increases that Amazon matches within 48 hours with rival increases that Amazon does not match within 48 hours. Panel (b) makes the analogous comparison for rival decreases. Lines report 5-day centered moving averages of daily persistence shares. For rival increases, persistence means that the rival's later price remains at least as high as the new price set immediately after the increase. For rival decreases, persistence means that the rival's later price remains no higher than the new price set immediately after the decrease. For example, a \$100-to-\$110 increase is counted as persistent only if the later rival price is at least \$110; a \$100-to-\$90 decrease is counted as persistent only if the later rival price is no higher than \$90.

The raw checkpoint values show the same pattern. At two weeks, 67.1% of matched rival increases remain at or above the new price, compared with 51.3% of unmatched increases; the gap remains at 60 days (63.0% versus 47.1%). Product-clustered checkpoint tests show that the increase gap is statistically significant at two weeks (15.8 percentage points, $SE = 5.8$, $p = .007$) and remains positive at 60 days (15.9 percentage points, $SE = 8.3$, $p = .058$). Rival decreases show the opposite pattern: matched decreases persist less often than unmatched decreases at two weeks

(44.6% versus 49.2%), though this gap narrows by 60 days (37.7% versus 38.6%). These decrease gaps are not statistically significant (-4.5 percentage points, $SE = 5.0$, $p = .361$ at two weeks; -0.9 percentage points, $SE = 7.2$, $p = .895$ at 60 days). Amazon's matching response is therefore associated with more durable higher rival prices specifically, not with durable rival price changes in general.

Taken together, the evidence identifies the empirical pattern that the model needs to explain. Despite its large market position, Amazon is rarely the high-price platform. Its distinctive role instead appears in the sequence of responses: Amazon often raises price after isolated rival increases, and its responses frequently preserve the rival's higher price rather than undoing it through deep undercutting. Those rival increases also last longer when Amazon matches them. The next section builds a model to explain why following can be incentive-compatible for a demand-advantaged firm.

3 Dominant Followership

When two undifferentiated firms compete only on price, the standard benchmark is the simultaneous-move Bertrand model, which predicts prices equal to marginal cost. Sequential timing alone does not overturn this intuition: if firms are otherwise symmetric and can undercut each other, the force pushing prices toward marginal cost remains. The standard intuition changes once the model is modified in a seemingly minor way: one firm has a demand advantage at price parity. Instead of assuming that firms are identical in every respect, I allow one firm, hereafter the dominant firm, to capture more than half of market demand when prices are equal. This assumption can be microfounded by a simple lexicographic decision rule. Consumers first compare prices, but when prices are equal or perceived as equivalent, they choose based on firm identity, brand, convenience, or other non-price advantages that favor the dominant firm. This advantage need not operate through the dominant firm charging high prices itself or moving first to claim the usual Stackelberg advantage: the relevant margin is the follower's response incentive when a rival moves first.

The analysis builds the mechanism in steps. I first isolate the core mechanism in a one-clearing pricing game: demand advantage raises prices only when the demand-advantaged firm is the follower, since only then does the advantage turn the follower's incentive from undercutting to matching. Appendix B.6 shows that this mechanism survives when firms retain loyal consumers at higher prices. I then add an interim market clearing, introducing a dynamic tradeoff between the short-run cost of initiating a price increase and the future value of the dominant firm's response. This tradeoff depends on how reliably the dominant firm can be counted on to adjust its price after the rival adjusts its own. The infinite-horizon extension asks whether the resulting high price persists once the game continues, rather than only for the one extra period a finite game can capture. Once the high matched price is reached, the rival can always undercut it; what deters this ongoing temptation is the rival's own response reliability.

3.1 Environment

Suppose there are two firms, indexed by $i \in \{D, S\}$. Firm D is the dominant, demand-advantaged firm, and firm S is the less demand-advantaged rival. Firms choose prices from a finite evenly spaced grid

$$\mathcal{P} = \{\delta, 2\delta, \ldots, M\delta\},$$

where $\delta > 0$ is the grid step and $M\delta$ is the maximum price. Let $p_i \in \mathcal{P}$ denote firm i's price. For any x, define the grid floor

$$\lfloor x \rfloor_{\mathcal{P}} = \max\{p \in \mathcal{P} : p \leq x\}.$$

For numerical illustrations, I normalize the maximum price to one and use a five-point grid, so $\delta = 0.2$ and $\mathcal{P} = \{0.2, 0.4, 0.6, 0.8, 1\}$.

The grid reflects the discrete price changes observed in the data.[8]

Total demand is normalized to one. Demand is allocated according to

$$q_D(p_D, p_S) = \begin{cases} 1, & p_D < p_S, \\ \lambda, & p_D = p_S, \\ 0, & p_D > p_S, \end{cases} \qquad q_S(p_D, p_S) = 1 - q_D(p_D, p_S).$$

[8] It can also be interpreted as a reduced-form representation of limited consumer price perception: under Weber's law, the smallest noticeable change in a stimulus is proportional to its initial level, so sufficiently small relative price differences may generate the same perceived price and demand response (Fechner 1860; Monroe 1971, 1973). Observed pricing patterns are consistent with discrete, economically meaningful price steps: average dollar adjustment sizes are \$5.17 for products under \$50, \$13.08 for products between \$50 and \$150, \$28.55 for products between \$150 and \$500, and \$81.64 for products above \$500.

The low-price firm receives the entire market. When prices are equal, the dominant firm receives share $\lambda \in [1/2, 1]$, with $\lambda = 1/2$ corresponding to the symmetric benchmark. I treat λ as capturing non-price platform advantages, such as brand trust, membership programs, or switching frictions, rather than an advantage created within the model by current-period pricing.

Timing and market clearing. The baseline analysis compares two standard one-shot timing benchmarks: simultaneous-move Bertrand competition, in which firms set prices without observing each other's current choice, and sequential-move Bertrand competition (Stackelberg timing), in which one firm, the leader, sets its price first and the other, the follower, observes it before responding. In both cases, the market clears only once, after both firms have set their prices, so there is no demand or payoff before the full price pair is determined. This one-shot structure means the game ends with zero probability of either firm getting a further opportunity to respond. With marginal cost normalized to zero, realized profit is

$$\pi_i(p_D, p_S) = p_i q_i(p_D, p_S).$$

The two-clearing and infinite-horizon extensions below relax this zero-response-probability assumption, introducing timing friction as a force distinct from demand asymmetry.

3.2 One-Clearing Pricing Games

Matching versus undercutting. The key mechanism can be seen from a firm's best response to a rival price $p \in \mathcal{P}$. On the price grid, the relevant local choice is whether to match p or undercut it by one step, $p - \delta$, since any lower price also receives the entire market but yields a lower margin. If the firm receives demand share s at price parity, matching yields sp, while undercutting yields $p - \delta$. The firm is therefore willing to match if and only if

$$sp \geq p - \delta,$$

or equivalently,

$$p \leq \frac{\delta}{1 - s}. \tag{1}$$

The matching cutoff $\delta/(1-s)$ is increasing in the firm's demand share at parity.

Simultaneous pricing. Consider a candidate common-price equilibrium $p_D = p_S = p$ under simultaneous pricing. Applying condition (1), the dominant firm's no-undercutting condition is

$$p \leq \frac{\delta}{1-\lambda},$$

while the rival's condition is

$$p \leq \frac{\delta}{\lambda}.$$

Under demand asymmetry, $\lambda > 1/2$, the rival's condition is more restrictive. Hence the highest common-price equilibrium is

$$p^{Sim}(\lambda) = \left\lfloor \frac{\delta}{\lambda} \right\rfloor_{\mathcal{P}}.$$

Thus, demand asymmetry by itself is not enough to raise prices in the simultaneous benchmark. The common price is disciplined by the rival, which has the stronger incentive to undercut.

Sequential pricing. Sequential pricing changes the constraint because the follower observes the first mover's price before choosing. This one-clearing timing structure gives the first mover no opportunity to revise again before demand is realized. The highest sustainable matched price is therefore pinned down by the follower's willingness to match.

Let p_D^{follow} denote the highest matched price sustainable when the dominant firm follows, and let p_S^{follow} denote the corresponding price when the rival follows.

Proposition 1 (Dominant followership raises equilibrium prices)**.** *Suppose $\lambda > 1/2$. The highest matched price sustainable under sequential pricing satisfies*

$$p_D^{follow} \geq p_S^{follow}.$$

[9] *Specifically,*

$$p_D^{follow} = \left\lfloor \frac{\delta}{1-\lambda} \right\rfloor_{\mathcal{P}},$$

[9] The inequality is weak because prices are chosen from a discrete grid; it is strict whenever $\mathcal{P}$ contains a price in $(\delta/\lambda, \delta/(1-\lambda)]$.

while

$$p_S^{follow} = \left\lfloor \frac{\delta}{\lambda} \right\rfloor_{\mathcal{P}}.$$

The result follows from applying the no-undercutting condition (1) to the follower: $\lambda > 1/2$ implies $\delta/(1-\lambda) > \delta/\lambda$. Figure A.2 plots the comparative statics.

Table 3 illustrates the result in a five-point price grid.

Table 3. Matched Equilibrium Prices: An Illustration

Demand environment	Simultaneous pricing	Sequential $D \to S$	Sequential $S \to D$
Symmetric demand, $\lambda = 1/2$	2δ	2δ	2δ
Asymmetric demand, $\lambda = 0.7$	δ	δ	3δ

Notes: Entries report the highest matched equilibrium price on the normalized five-point grid. In sequential pricing, $D \to S$ denotes the dominant firm moving first and the rival following; $S \to D$ denotes the reverse.

The table shows that equilibrium prices rise only when the two forces coincide. Demand asymmetry alone is not enough: under simultaneous pricing or when the dominant firm leads ($D \to S$), the rival's no-undercutting condition binds either way, so the price is no higher than under symmetric demand. Sequential timing alone is not enough either: without demand asymmetry, all three timing structures give the same matched price. The price rises only when asymmetric demand is paired with the dominant firm following ($S \to D$). In that case, the same price increase creates opposite response incentives: the dominant firm is willing to match because it keeps substantial demand at comparable prices, while the rival has a stronger incentive to undercut. Appendix B.2 formalizes this behavioral separation and shows that both firms can prefer the dominant-followership role assignment among matched sequential outcomes.

3.3 Costly Rival Initiation

The one-clearing game abstracts from the current cost of initiating a price increase: if the dominant firm matches, demand is realized only after both firms are at the higher price. I next ask whether dominant followership can still support rival initiation when initiation is costly. I introduce this cost by giving the dominant firm one extra chance to revise its price, but only after the market has already cleared once. Because the market clears before that final revision, the rival may lose current demand before the dominant firm matches. The key question is whether the future value of the dominant firm's match is large enough to offset the rival's current demand loss.

I write this timing as $i \to j \to M \to i \to M$, where M denotes a market clearing. Let a_1 denote firm i's initial price, a_2 firm j's response, and a_3 firm i's final price. The two market clearings occur at price pairs (a_1, a_2) and (a_3, a_2), so firm k's total payoff is

$$U_k(a_1, a_2, a_3) = \pi_k(a_1, a_2) + \beta \pi_k(a_3, a_2), \qquad k \in \{i, j\}, \tag{2}$$

where $\beta \in (0, 1]$ weights the second market. I solve this finite game by backward induction.

Proposition 2 (A threshold for costly rival initiation)**.** *Consider the order $D \to S \to M \to D \to M$. Fix the dominant firm's initial price at δ[10], and let $p^M \equiv \lfloor \delta/(1-\lambda) \rfloor_{\mathcal{P}}$ denote the highest price the dominant firm will match at the final revision. Whenever $p^M > \delta$, the rival's best response is to raise price to p^M if and only if*

$$\beta > \bar{\beta}(\lambda) \equiv \frac{\delta}{p^M - \delta}.$$

The threshold $\bar{\beta}(\lambda)$ is weakly decreasing in λ: a larger demand share at parity requires less patience to make rival initiation worthwhile.

Corollary 4 in Appendix B.3 verifies that this threshold is strong enough to support a complete equilibrium, not only a conditional response rule.

The result shows when costly initiation can occur. The rival is willing to raise price first only when it is sufficiently forward-looking: it must value the future matched price enough to compensate for the current demand it gives up. The firm that initiates the increase need not be the firm that makes it sustainable. The identity of the final responder is therefore essential: if the rival receives the final revision opportunity instead, its stronger incentive to undercut prevents the high price from being sustained. Appendix B.4 provides this reverse-timing comparison.

Figure 5 plots the threshold directly. As λ rises, the dominant firm becomes willing to match a higher price, $p^M(\lambda)$. A higher p^M raises the rival's payoff after the dominant firm matches, so

[10] The grid's lowest price is a natural starting point for this illustration because it is the empirically relevant case: Amazon is typically at or near the lowest price among competitors (Figure 1: tied with the lowest rival in 54.2% of comparisons, priced above it in only 14.4%), though the grid floor here is a modeling device and not a claim that some absolute lowest price exists in practice.

initiation is profitable at a lower discount factor.

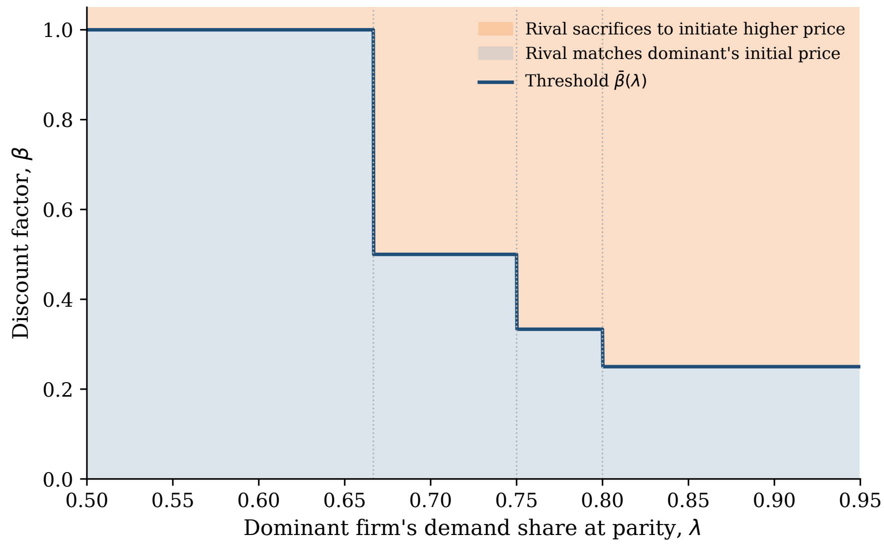

Figure 5. Threshold for Rival Price Initiation.

Notes: The figure uses the normalized five-point grid for the $D \to S \to M \to D \to M$ timing in Proposition 2, fixing the dominant firm's initial price at δ. The line plots $\bar{\beta}(\lambda) = \delta/(p^M(\lambda) - \delta)$, the exact threshold above which the rival's unique best response is to raise price to $p^M(\lambda)$ rather than match δ. Dotted vertical lines mark the grid points at which $p^M(\lambda)$ steps up.

Proposition 2 assumes that the dominant firm's final response is guaranteed. I next relax this by introducing response reliability. Let $\rho_i \in [0, 1]$ denote the probability that firm i can revise its price when its turn immediately follows a rival price change. If the rival has not changed price, the firm can revise normally, without ρ_i entering. In the two-clearing game, the dominant firm's final turn is therefore gated only if the rival moves away from the shared starting price, $a_2 \neq a_1 = \delta$.[11] With probability ρ_D, the dominant firm revises optimally; with probability $1 - \rho_D$, it does not revise and its price remains at δ.

Corollary 1 (Response reliability and costly initiation)**.** *Under the timing of Proposition 2, suppose the dominant firm's final turn is gated only after a rival price change. Then the rival's threshold for costly initiation is*

$$\bar{\beta}(\lambda, \rho_D) \equiv \frac{\delta}{\rho_D\, p^M(\lambda) - \delta},$$

[11] In this finite game, $a_2 \neq a_1$ is the analogue of a rival price change. The infinite-horizon game and the learning simulations use the same response-window rule: the window is triggered when the rival changes its posted price relative to its previous posted price.

which equals $\bar{\beta}(\lambda)$ when $\rho_D = 1$. If $\rho_D \, p^M(\lambda) \leq \delta$, no $\beta < 1$ satisfies this threshold, and costly initiation cannot be rationalized at any discount factor.

Patience and response reliability are therefore complements. The rival is willing to sacrifice current demand only if it is patient enough and the dominant firm is reliable enough to make matching likely; a sufficiently unreliable dominant firm forecloses costly initiation regardless of how patient the rival is.

3.4 Infinite-Horizon Markov Extension

The finite-game extension asks whether a rival can be willing to initiate a higher price when doing so is costly in the current period. I next ask whether the resulting high matched price can persist once the game continues. The new issue is defection: after the market reaches the high matched price, the rival may undercut whenever it gets another chance to move.

Consider the two-price case $\mathcal{P} = \{L, H\}$, where $0 < L < H$. Firms alternate revision opportunities on a fixed clock, D then S then D then S. I use the same response-window rule as above. A firm moves normally unless its turn immediately follows a rival price change. In that case, it can revise only with probability ρ_i; otherwise prices remain unchanged for that period. The Markov state is (p_D, p_S, m, c): the current posted-price pair, the firm $m \in \{D, S\}$ whose turn is next, and an indicator c for whether the preceding period involved a price change. For the analytical result, I fix $\rho_D = 1$, so the dominant firm is fully reliable, and vary ρ_S. This isolates the rival's own response reliability: how quickly it can re-establish a high price after giving one up.

Section 3.3 showed why high ρ_D matters for initiation: the rival is willing to raise price only if the dominant firm's later match is likely enough. I therefore set $\rho_D = 1$ and verify a candidate policy in which the rival initiates the high price from (L, L), the dominant firm matches, and both firms remain at (H, H):

price state (p_D, p_S)	$\sigma_D(p_D, p_S)$	$\sigma_S(p_D, p_S)$
(H, H)	H	H
(H, L)	L	H
(L, H)	H	H
(L, L)	L	H

where $\sigma_i(p_D, p_S)$ is firm i's price choice when it receives the revision opportunity.

The relevant price region is the same matching-versus-undercutting wedge as Section 3.2: the dominant firm is willing to match H, while the rival would undercut H if it could immediately exploit the matched price. Appendix B.5 verifies that the rival prefers preserving H to undercutting it, along with the initiation and off-path checks, by solving the Bellman equations.

Proposition 3 (Dominant followership can persist in repeated competition)**.** *Suppose $\lambda > 1/2$, $\rho_D = 1$, and*

$$(1 - \lambda)H < L < \lambda H.$$

There is a nonempty parameter region in which a Markov-perfect equilibrium has the rival initiate the high price, the dominant firm match, and both firms remain at (H, H) indefinitely. This region weakly expands as the rival's response reliability ρ_S falls.

The new force in the repeated game is the rival's temptation to undercut once the market reaches (H, H). Lower ρ_S makes this temptation easier to deter: after undercutting, the rival may have to wait before raising price back to H. Figure 6 illustrates this logic. After defecting, a moderately reliable rival ($\rho_S = 0.5$) returns to H faster than a fully unreliable rival ($\rho_S = 0$), holding $\rho_D = 1$ fixed throughout as in the proposition. The shaded gap is the rival's own cost of undercutting.

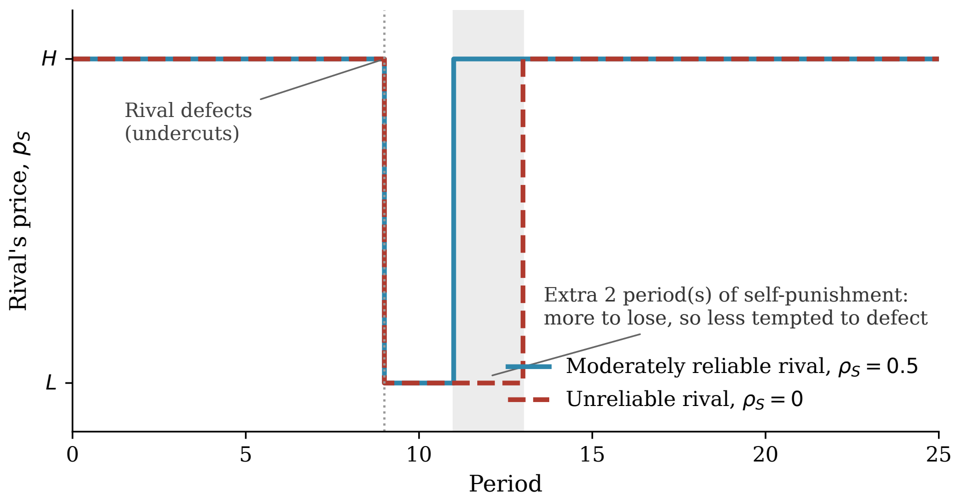

Figure 6. A Rival's Defection Costs Itself More When It Is Unreliable

Notes: Simulated path of the rival's own price under the candidate policy and the response-window rule in Section 3.4, with $\rho_D = 1$ throughout and the same sequence of underlying random draws for both lines, so only ρ_S differs between them. The rival is forced to defect at period 9; play afterward follows the candidate policy and the stochastic response-window rule with no further intervention. This illustrates the mechanism behind Proposition 3 rather than proving it; the incentive conditions themselves are verified directly in Appendix B.5.

The matching pattern from Section 3.3 therefore does not unravel once the game keeps going. Lower ρ_S makes undercutting less attractive for the rival, making the high price easier to sustain. Together, Corollary 1 and Proposition 3 motivate the asymmetric-response treatment in Section 4: $\rho_D = 1$ makes costly initiation more attractive, while $\rho_S = 0$ makes the matched price more persistent.

4 Algorithmic Learning with Dominant Followership

The model assumes that firms choose best responses with knowledge of the payoff environment. In online retail, that is a strong assumption. Firms set prices across large product assortments, while demand conditions and rival response patterns vary across products and change in real time. A pricing algorithm may therefore have to learn the mechanism from market feedback rather than solve it from a known model. I ask whether Q-learning algorithms can do so. The canonical Q-learning literature studies symmetric firms that move simultaneously (Calvano, Calzolari, Denicolò, and Pastorello 2020; Banchio and Mantegazza 2023; Asker, Fershtman, and Pakes 2024); Klein (2021) extends this to sequential, alternating-move pricing, but firms remain symmetric in their opportunity to respond. I depart from both benchmarks by allowing firms to differ in demand advantage and in how reliably they can respond, while still setting prices sequentially. This gives the simulations the two features missing from the existing Q-learning literature but central to the empirical setting: demand asymmetry and asymmetric response reliability.

The results then proceed in three steps. I first ask when learned prices rise and show that demand asymmetry raises prices only when it is paired with asymmetric response timing. I then use one-learner decompositions to identify whose learning sustains the price increase. Finally, I show that the learning that raises prices and the gains from those prices need not accrue to the same firm.

4.1 Environment and Learning Rule

Environment. I embed the demand environment from Section 3 in a repeated learning problem with two firms, D (demand-advantaged) and S (the rival), each choosing prices from a common grid $\mathcal{P}$. At each pricing decision, firm i observes the inherited posted-price pair $s_{i,t} = (p_{i,t-1}, p_{j,t-1})$, $j \neq i$, and chooses a new price; Appendix Section A.2 shows this two-price state is well supported empirically. Once both firms' posted prices are set for the period, each earns the realized profit $\pi_i(p_{i,t}, p_{j,t})$ from the demand environment in Section 3.1.

Timing. I compare two timing structures. In a simultaneous-updating benchmark, both firms choose prices before demand is realized, so neither observes or responds to the rival's current move; this shuts down the response channel and serves as a no-response benchmark. In posted-price response updating, at most one firm adjusts its price each period, demand is realized after that move, and the rival may receive a response opportunity in the following period. Whether it does is stochastic: ρ_j matters only after firm i changes its price. After such a price change, firm j can revise on its next scheduled turn with probability ρ_j. If the response window does not activate, no firm moves in that period, and demand is realized again at unchanged prices. This is the same response-window structure analyzed in Section 3.4. The Q-learning simulations replace firms' knowledge of the payoff environment with learning from realized profit feedback under this timing rule. Letting ρ_D and ρ_S differ lets the simulations reflect the asymmetric response timing documented empirically, rather than treating algorithmic competition as symmetric and simultaneous. Appendix Section C.1

gives the exact response-opportunity rule.

Learning rule. Firms use standard Q-learning with ϵ-greedy exploration: each firm maintains an estimate $Q_{i,t}(s,a)$ of the discounted value of price a in state s, chooses its estimated best price with probability $1-\epsilon_t$ and a random price otherwise, and updates its estimate toward a realized target $\widehat{Q}_{i,t}$ using learning rate α_t:

$$Q_{i,t+1}(s_{i,t},a_{i,t}) \leftarrow (1-\alpha_t)Q_{i,t}(s_{i,t},a_{i,t}) + \alpha_t\widehat{Q}_{i,t}(s_{i,t},a_{i,t}).$$

The only difference between the two timing structures is what that target credits. In the simultaneous benchmark, both firms choose before demand is realized, so the target is one period of profit:

$$\widehat{Q}^{sim}_{i,t} = r_{i,t} + \beta \max_{a'\in\mathcal{P}} Q_{i,t}(s_{i,t+1},a').$$

In posted-price response updating, firm i's price remains active until its next pricing decision, so the target credits the full profit path realized while it is posted, including profit earned after a possible rival response, plus continuation value from the resulting state. In the usual two-turn case,

$$\widehat{Q}^{post}_{i,t} = \pi_i(p_{i,t},p_{j,t}) + \beta\pi_i(p_{i,t+1},p_{j,t+1}) + \beta^2 \max_{a'\in\mathcal{P}} Q_{i,t}(s_{i,t+2},a'). \tag{3}$$

Appendix Section C.1 gives the exploration schedule and the general target for when a scheduled response window goes unused.

4.2 Simulation Design

Main simulation treatments. The main simulation treatments use the repeated response-window timing from Section 3.4. A response opportunity arises after a rival price change, and firms can differ in how reliably they receive those opportunities. The simulations therefore replace firms' knowledge of the payoff environment with learning from realized profit feedback while keeping the timing mechanism fixed.

The treatments vary demand and response timing. Table 4 summarizes the resulting 2×2 design.

Table 4. Simulation Treatments

	Response timing	
Demand environment	Symmetric response	Asymmetric response
Symmetric demand	$\lambda = 0.5,\ (\rho_D, \rho_S) = (0.5, 0.5)$	$\lambda = 0.5,\ (\rho_D, \rho_S) = (1, 0)$
Asymmetric demand	$\lambda = 0.7,\ (\rho_D, \rho_S) = (0.5, 0.5)$	$\lambda = 0.7,\ (\rho_D, \rho_S) = (1, 0)$

Notes: The table reports the four main two-learner simulation treatments. Symmetric demand sets $\lambda = 0.5$; asymmetric demand sets $\lambda = 0.7$. Symmetric response sets equal response probabilities; asymmetric response makes the dominant firm the reliable responder after a rival price change.

A separate simultaneous-updating benchmark shuts down the response channel entirely and compares $\lambda = 0.5$ with $\lambda = 0.7$, holding the price grid and learning rule fixed. This benchmark isolates the demand effect before introducing symmetric or asymmetric response timing.

Two-learner simulations. The main simulations let both firms use Q-learning pricing algorithms. These runs provide the primary comparison across the four cells in Table 4, allowing demand asymmetry and response timing to interact while holding the learning rule fixed.

One-learner decomposition. The two-learner simulations combine two forces: each firm learns, and each firm responds to the other's learned prices. To separate these forces, I also run one-learner benchmarks in which one firm uses Q-learning and the other follows a myopic best response: the price that maximizes its current-period profit against the rival's last posted price, without updating continuation values or internalizing how the choice affects future states. Comparing the cases in which only D learns, only S learns, and both firms learn shows whether the learned price increase is driven by the dominant firm's own learning, by the rival learning from the dominant firm's responses, or by their interaction.

Outcome measurement. The main outcome is the transaction price, $p_t^{mkt} = \min\{p_{D,t}, p_{S,t}\}$, the price paid by consumers under the baseline demand allocation. I also report firm-level posted prices and profits to show how learning changes the distribution of gains. Unless otherwise stated, outcomes are averaged over the final 5 percent of periods across simulation runs.

4.3 When Do Learned Prices Rise?

Simultaneous no-response benchmark. The simultaneous-updating benchmark removes the response channel and isolates the demand effect. The learned transaction price is 0.339 under symmetric demand and 0.274 under asymmetric demand. Thus, without a response channel, demand asymmetry lowers rather than raises learned prices. This benchmark shows that, even in repeated learning, demand advantage by itself does not generate higher prices when firms cannot respond to each other's current price changes. Appendix Figure A.4 shows the corresponding price trajectories.

Appendix C.3 reports a complementary one-clearing sequential benchmark that preserves leader-follower roles but removes the posted-price response channel. This benchmark shows that the role asymmetry in the static model also appears under learning, while the main results below add the posted-price feedback through which current prices affect both payoffs and future states.

Table 5. Demand Advantage, Response Timing, and Learned Pricing

	Response timing	
	Symmetric response	Asymmetric response
Panel A. Transaction price		
Symmetric demand	0.388	0.533
Asymmetric demand	0.433	0.660
Panel B. Posted prices, (p_D, p_S)		
Symmetric demand	(0.457, 0.450)	(0.550, 0.680)
Asymmetric demand	(0.490, 0.563)	(0.667, 0.776)
Panel C. Firm profits, (π_D, π_S)		
Symmetric demand	(0.192, 0.196)	(0.403, 0.129)
Asymmetric demand	(0.288, 0.145)	(0.545, 0.116)

Notes: The table reports averages over the final 5 percent of periods in the two-learner Q-learning simulations. The transaction price is $p_t^{mkt} = \min\{p_{D,t}, p_{S,t}\}$, reflecting that demand is allocated to the lower-priced firm, with demand split or assigned by λ at price parity. In ordered pairs, D is the dominant firm and S is the less demand-advantaged rival. Symmetric demand sets $\lambda = 0.5$; asymmetric demand sets $\lambda = 0.7$. Symmetric response sets $\rho_D = \rho_S = 0.5$; asymmetric response sets $\rho_D = 1$ and $\rho_S = 0$.

Learned prices under dominant followership. Table 5 shows that the highest prices arise when demand advantage and asymmetric response timing operate together. Three comparisons are useful. First, demand asymmetry alone does little to raise learned prices: under symmetric response timing, the market price rises only from 0.388 to 0.433. Second, response timing matters because it changes the payoff from initiating a price increase: when the dominant firm subsequently prices close to the rival rather than pushing the market back to low prices, the market can remain in a high-price state. Anticipating this response makes initiation more attractive because the rival is not left alone at the higher price for as long. Third, the largest price increase occurs under dominant followership, when the demand-advantaged firm is the reliable responder. In that case, the learned market price reaches 0.660. The firm-specific prices and profits preview the distributional result below: the rival posts the higher price on average, while the dominant firm earns most of the profit. This pattern also appears at the run level. Appendix C.4 shows that this relationship is consistent across simulation runs: runs with higher rival posted prices also tend to have higher dominant-firm profits. In these runs, the rival is often the firm posting higher prices, but the dominant firm earns more from the resulting high-price states.

Reinforcement from matching responses. The key learning object is whether a higher-price attempt is rewarded or punished by the rival's response. Figure 7 measures this feedback by asking whether a firm that posts a higher price is matched by the firm with the next response opportunity. In asymmetric-demand cells, this asks whether the rival's higher-price move is matched by the demand-advantaged firm.

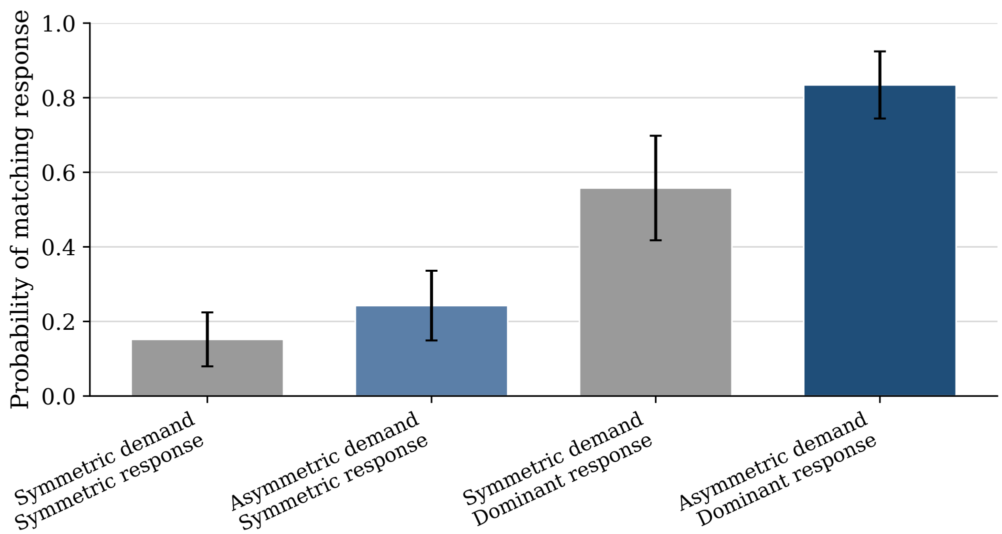

Figure 7. Matching Responses after Rival Higher Prices.

Notes: Bars report the mean, across simulation runs, of the probability that the response-side firm matches after the other firm posts above its inherited posted price, using events in the final 5 percent of periods. The response-side firm is the firm whose response probability is increased in the asymmetric-response treatment; in asymmetric-demand cells, it is the demand-advantaged firm. Whiskers report 95 percent confidence intervals across simulation runs.

The figure shows that matching becomes much more likely when the two asymmetries operate together. Demand asymmetry alone raises the matching probability modestly, from 0.15 to 0.24 under symmetric response. Response timing alone has a larger effect, raising it from 0.15 to 0.56 under symmetric demand. The largest effect occurs when the reliable responder is also demand advantaged: the matching probability reaches 0.83. This is the learning counterpart of dominant followership. The dominant firm's response changes the feedback attached to the rival's price increase, making higher-price attempts more likely to be reinforced rather than punished. Appendix Figure A.6 decomposes the nonmatching responses.

Concentration in high-price states. Figure 8 shows where the algorithms spend time after learning. For each simulation treatment, the figure reports the empirical distribution of posted-price pairs in the final part of training, pooling observations across simulation runs.

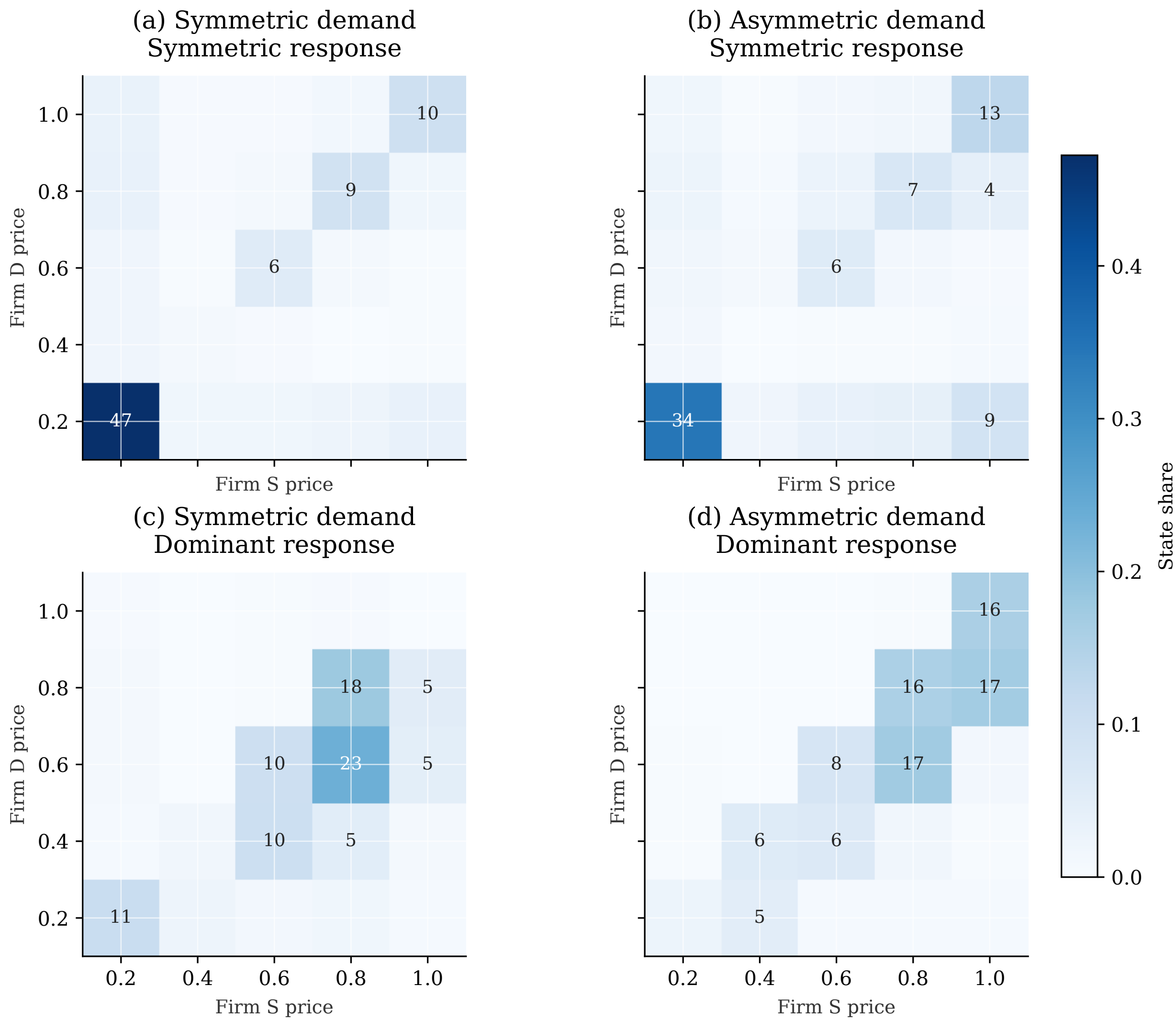

Figure 8. Visited Price States after Learning.

Notes: For each scenario, the figure pools posted-price pairs across all simulation runs and all periods in the final 5 percent of training, then reports the share of pooled observations in each price state. The vertical axis is firm D's posted price and the horizontal axis is firm S's posted price. In asymmetric-demand panels, D is the demand-advantaged firm and S is the rival; in symmetric-demand panels, the same labels are kept only to make the state space comparable across panels. Cell labels report percentage shares and are shown for cells with shares of at least 4 percent.

The heatmaps show that the high average prices in Table 5 are not driven by isolated price spikes. In the symmetric-demand, symmetric-response benchmark, nearly half of late-period observations remain at the lowest price pair. Under dominant followership, by contrast, the mass of the distribution moves toward the upper-right part of the state space, where both firms post high prices. The response feedback documented above is therefore reflected in the states visited after learning: higher-price attempts that are repeatedly matched are associated with late-period play in which both firms often post high prices. Appendix Figure A.7 shows that movement into these high-price states occurs during training, not only in the final cross-section.

4.4 Whose Learning Sustains Higher Prices?

The role of rival learning. The two-learner price increase could arise because the dominant firm learns to use its demand advantage, or because the rival learns that higher prices are likely to be accommodated by the dominant firm's response. Table 6 separates these channels by holding the dominant-followership environment fixed and varying only Q-learning adoption. The contrast is sharp. When only the dominant firm learns, the market price is 0.225. When only the less demand-advantaged rival learns, the market price rises to 0.589. When both firms learn, it rises further to 0.660.

Table 6. Q-Learning Adoption in the Dominant-Followership Environment

Q-learning adoption	Market price	Dominant posted price	Rival posted price
Dominant firm only	0.225	0.248	0.232
Less demand-advantaged rival only	0.589	0.592	0.598
Both firms	0.660	0.667	0.776

Notes: The table reports averages over the final 5 percent of periods across 30 simulation runs. The environment is held fixed at the dominant-followership treatment: asymmetric demand, with $\lambda = 0.7$, and asymmetric response timing, with $\rho_D = 1$ and $\rho_S = 0$. When only one firm uses Q-learning, the other firm follows a myopic current-period best response. The market price is the minimum posted price.

The decomposition points to the rival's pricing decision. The market price can rise only if the rival is willing to post higher prices. When the rival is myopic, it keeps choosing the current best response to the dominant firm's posted price, so the market price remains low. It does not learn that a higher price today may be followed by a matching response tomorrow. When the rival uses Q-learning, it can learn this continuation value. The dominant firm does not need to learn the initiating role itself: because it is demand advantaged, matching a higher rival price can already be attractive as a current-profit best response. The key learning channel is therefore the rival learning that higher prices can be profitable when the dominant firm predictably accommodates them.

4.5 Who Captures the Gains?

Distribution of learning gains. Figure 9 asks who gains from the rival learning to initiate higher prices. The comparison holds the dominant firm fixed as a Q-learning responder and changes the rival from a myopic best response to Q-learning. The market price rises by 0.435, from 0.225 to 0.660. Both firms' profits increase, but the gains are highly asymmetric: the dominant firm's profit rises by 0.397, compared with 0.039 for the rival. The learning that raises prices and the gains from those higher prices therefore need not accrue to the same firm. A rival's learning algorithm can help sustain higher prices, while the dominant firm receives most of the resulting surplus because of its demand advantage.

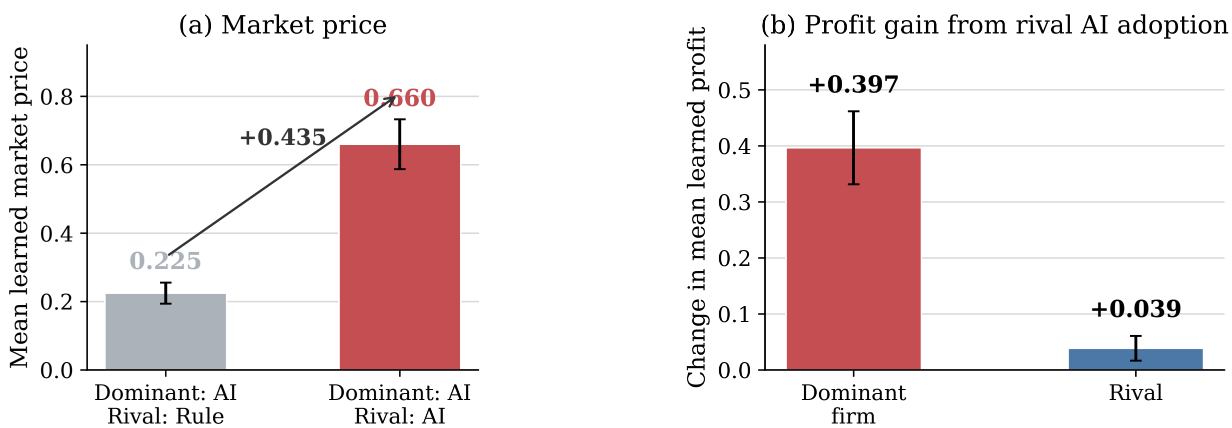

Figure 9. Market-Price and Profit Effects of Rival Q-Learning Adoption.

Notes: The figure uses the asymmetric-demand, asymmetric-response environment. It compares the benchmark in which the dominant firm uses Q-learning and the rival uses a myopic best response with the two-learner environment in which both firms use Q-learning. Panel (a) reports the change in the market price. Panel (b) reports changes in firm profits. Whiskers report 95 percent confidence intervals across 30 simulation runs. All outcomes are averaged over the final 5 percent of simulation periods.

Taken together, the simulations show how dominant followership operates when prices are set by learning algorithms. Demand asymmetry alone does not raise learned prices. Prices rise only when the demand-advantaged firm is also the reliable responder. In that case, the dominant firm's matching response makes the rival's higher-price experiments more rewarding to learn, and repeated learning moves the market toward high-price states. The dominant firm does not need to learn the price increase itself. Its role is to make the rival's price increase profitable to learn, while its demand advantage allows it to capture most of the gains.

5 Conclusion

This paper shows why algorithmic retail competition cannot always be understood by asking which firm moves first. In high-frequency pricing environments, a firm's response can be the strategic margin that shapes the market outcome. When the demand-advantaged firm is also the reliable responder, it can raise prices by following.

The evidence, model, and simulations all point to this response channel. The empirical analysis makes the channel visible: Amazon often plays the responder role rather than the initiator role. The model explains why this role can be privately optimal for a demand-advantaged firm. The learning simulations show how this incentive operates when prices are set by adaptive algorithms: a rival's algorithm can learn to initiate higher prices, while the demand-advantaged firm captures most of the gains by responding.

The concern is not any single price response viewed in isolation. Nor does the mechanism require an agreement, communication, or a common pricing rule. Each response can be individually optimal given the firm's own demand position. The problem is that, once such responses become systematic and learnable, they can change what rival algorithms learn is profitable. A routine price response can therefore become part of a feedback system that softens competition.

This changes what needs to be measured and modeled in algorithmic pricing markets. Price levels alone are not enough. Regulators and researchers also need to know who responds after rival price changes, how reliably they respond, and whether the responder has a demand advantage. Counterfactual audits should ask whether market outcomes would change if response timing, response reliability, or demand advantage were different. Models that force pricing decisions into a symmetric simultaneous-move game miss the margin studied here. The relevant question is how asymmetric responses shape what algorithms learn and who benefits from the resulting market outcome.

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A Empirical Appendix

A.1 Response Robustness and Heterogeneity

Table A.1. Price Adjustment Responses to Rival Price Changes

	Panel A: Any rival move		Panel B: Isolated rival move	
	Price increase	Price decrease	Price increase	Price decrease
Amazon	15.90	8.91	12.94	12.78
	(3.19), $p < .001$	(3.30), $p = .007$	(1.79), $p < .001$	(1.71), $p < .001$
Walmart	0.82	1.93	4.04	1.30
	(2.11), $p = .697$	(1.96), $p = .324$	(1.86), $p = .030$	(2.02), $p = .521$
Target	3.41	2.37	4.42	4.07
	(1.42), $p = .017$	(1.66), $p = .154$	(1.34), $p = .001$	(1.22), $p = .001$
Best Buy	-0.83	-0.94	1.93	1.47
	(1.29), $p = .521$	(1.53), $p = .539$	(1.34), $p = .149$	(1.21), $p = .224$
Home Depot	4.25	4.37	6.12	8.30
	(5.18), $p = .412$	(4.40), $p = .321$	(2.56), $p = .017$	(3.22), $p = .010$
Product FE	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes
Day-of-week FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Clustered SE	Product	Product	Product	Product
Observations	597,309		597,334	

Notes: The unit of observation is platform-product-time. The responding platform is required not to change price at t. Panel A uses any rival price move for the same product at t. Panel B restricts to isolated rival price moves, in which no platform raised price for the product in the prior 48 hours, exactly one identified rival platform moves at t, and the responding platform's own price is unchanged at that moment. Coefficients are percentage-point effects. Standard errors, clustered by product, are reported in parentheses, followed by the exact two-sided p-value.

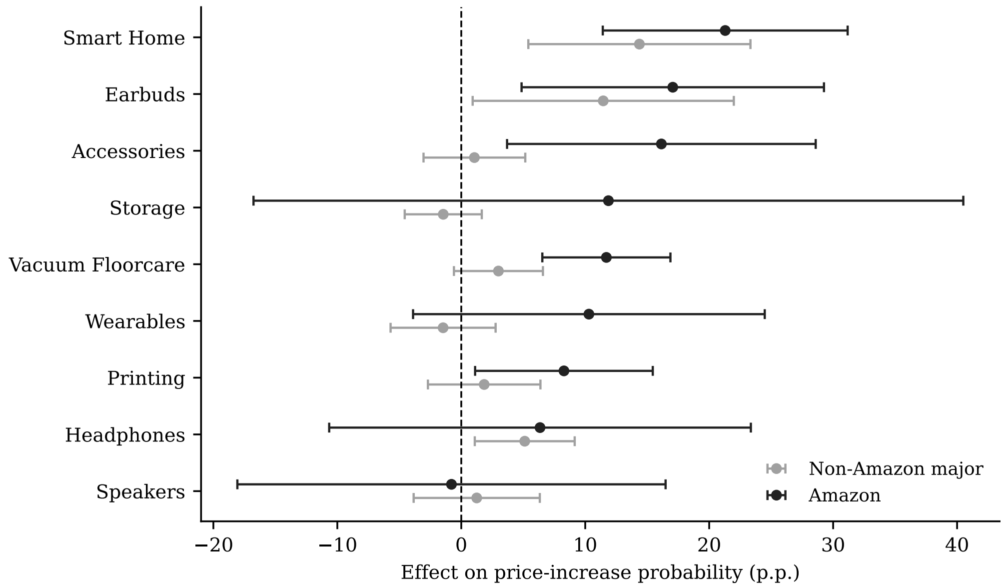

Figure A.1. Amazon Response to Isolated Rival Increases by Product Category.

Notes: The figure reports category-specific versions of the isolated-rival-increase response specification. Black markers report the Amazon coefficient; gray markers report the pooled coefficient for non-Amazon major platforms. Horizontal bars report 95 percent confidence intervals. Each specification includes product, week, day-of-week, and platform fixed effects, with standard errors clustered by product. The figure includes categories with at least 20 Amazon-facing isolated rival increase events and at least five product clusters.

Table A.2. Amazon Response to Isolated Rival Increases by Product Demand Tier

	Low demand (50–2K)	Middle demand (2K–8K)	High demand (9K–100K)
Amazon	8.28	10.94	15.50
	(5.02), $p = .099$	(2.75), $p < .001$	(2.55), $p < .001$
Non-Amazon major platforms	2.35	2.71	4.36
	(3.00), $p = .435$	(1.28), $p = .034$	(1.27), $p < .001$
Products	59	62	67
Observations	144,795	190,916	255,024

Notes: Demand tiers are terciles of each product's median Amazon "Bought in Past Month" demand proxy, computed over the full matched-product panel; the ranges shown are the approximate floor-coded bands from Amazon's own display and are not exact boundaries. Entries are percentage-point effects on the probability of increasing price within 48 hours of an isolated rival price increase, estimated jointly for Amazon and the pooled non-Amazon major platforms (Walmart, Target, Best Buy, and Home Depot) within each tier. Standard errors, clustered by product, are reported in parentheses, followed by the exact two-sided p-value. All specifications include product, week, day-of-week, and platform fixed effects.

A.2 Empirical State Dependence in Prices

The learning simulations use a parsimonious state: the platform's own lagged price and the lagged lowest rival price. Table A.3 supports this choice. Own prices are highly persistent, and the lagged lowest rival price helps explain the focal platform's current price even after conditioning on its own lagged price. The table also shows that a platform's position relative to the lowest rival price predicts the direction of its next adjustment: platforms priced above the rival benchmark tend to adjust downward, while platforms priced below it tend to adjust upward.

Table A.3. State-Space Motivation: Own and Rival Price States

	(1)	(2)	(3)
	$\log p_{ip,t}$	$\log p_{ip,t}$	$\Delta \log p_{ip,t}$
$\log p_{ip,t-1}$	0.9796***	0.9761***	
	(0.0046)	(0.0058)	
$\log p^{\min}_{-i,p,t-1}$		0.0080***	
		(0.0030)	
$\Delta \log p^{\min}_{-i,p,t-1}$			0.0105
			(0.0070)
$Gap_{ip,t-1}$			-0.0159***
			(0.0045)
Product FE	Yes	Yes	Yes
Week FE	Yes	Yes	Yes
Day-of-week FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Clustered SE	Product	Product	Product
Observations	878,992	878,992	849,328
R^2	0.999	0.999	0.008

Notes: The unit of observation is firm-product-time. The lag $t-1$ is the previous observed state for the same firm-product pair, restricted to observations between 1 and 6 hours apart. $p^{\min}_{-i,p,t}$ is the lowest rival price in the same product market, Δ denotes log price changes, and $Gap_{ip,t-1} = \log p_{ip,t-1} - \log p^{\min}_{-i,p,t-1}$. The sample restricts Amazon and Walmart to first-party listings where seller identity is observed. Standard errors are clustered by product. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B Theoretical Appendix

B.1 Comparative Statics for Sequential Matched Prices

Figure A.2 plots the highest sustainable matched price as a function of demand asymmetry and the price-grid increment.

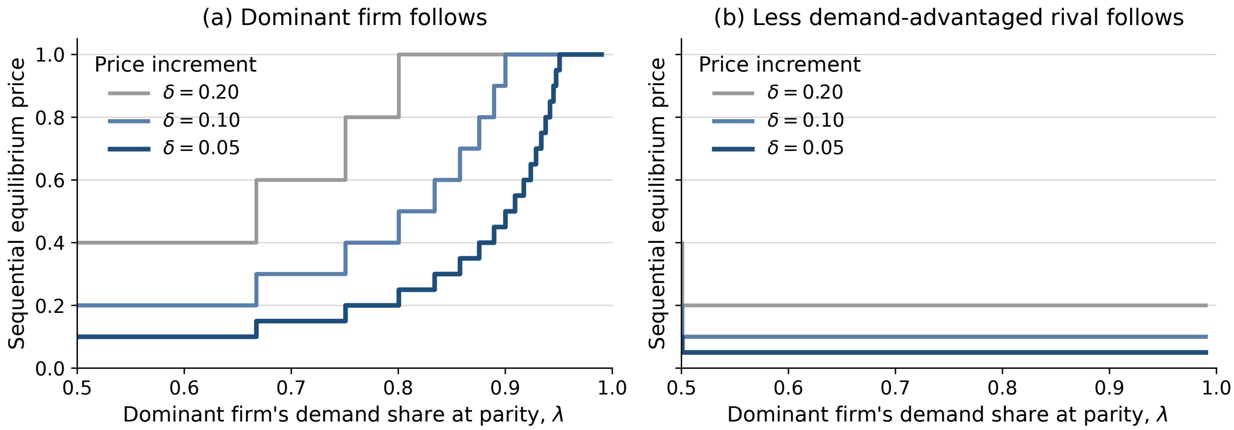

Figure A.2. Sustainable Matched Prices under Sequential Pricing.

Notes: Panel (a) shows the case in which the dominant firm follows, while Panel (b) shows the case in which the rival follows. Prices are the highest sustainable matched prices under the relevant follower constraint. Different lines correspond to different price-grid increments δ; smaller δ gives a finer price grid, so equilibrium prices adjust in smaller steps.

B.2 Behavioral Separation and Role Preference

Proposition 1 has two implications. First, there is a price region in which the dominant follower prefers matching while the rival follower prefers undercutting, so the same higher price can be sustained only when the dominant firm follows. Second, among matched sequential outcomes, both firms can prefer the role assignment in which the demand-advantaged firm follows.

Corollary 2 (Behavioral separation under demand asymmetry). *Suppose $\lambda > 1/2$. For any first-mover price $p \in \mathcal{P}$ satisfying*

$$\frac{\delta}{\lambda} < p < \frac{\delta}{1-\lambda},$$

the dominant firm matches as the follower, while the rival undercuts.

Let $\Delta_i^U(p)$ denote firm i's net gain, as follower, from undercutting a first-mover price p, relative to matching. For the dominant firm,

$$\Delta_D^U(p) = (1-\lambda)(p-\delta) - \lambda\delta = (1-\lambda)p - \delta.$$

For the rival,

$$\Delta_S^U(p) = \lambda(p-\delta) - (1-\lambda)\delta = \lambda p - \delta.$$

The incentives have opposite signs exactly when

$$\Delta_D^U(p) < 0 < \Delta_S^U(p) \quad \iff \quad \frac{\delta}{\lambda} < p < \frac{\delta}{1-\lambda}.$$

Corollary 3 (Role preference among matched outcomes)**.** *Among matched sequential outcomes, both firms weakly prefer $S \to D$ to $D \to S$.*

Let $S \to D$ denote the outcome in which the rival moves first and the dominant firm follows, and let $D \to S$ denote the reversed role assignment. Under $S \to D$, the matched price is p_D^{follow}; under $D \to S$, it is p_S^{follow}. At any matched price p, payoffs are

$$\Pi_D(p) = \lambda p, \qquad \Pi_S(p) = (1-\lambda)p.$$

Thus, $\Pi_i^{S\to D} = \Pi_i(p_D^{follow})$ and $\Pi_i^{D\to S} = \Pi_i(p_S^{follow})$. Since Proposition 1 gives $p_D^{follow} \geq p_S^{follow}$, both firms earn weakly higher payoffs under $S \to D$ among matched outcomes.

B.3 Proof of Proposition 2 and Corollaries 4, 1

Corollary 4 (A verified equilibrium)**.** *On the normalized five-point grid, for $\lambda \in (2/3, 3/4)$, $p^M = 3\delta$ and $\bar{\beta} = 1/2$. In this region, for $\beta > 1/2$, δ is also the dominant firm's own optimal initial price, so the unique equilibrium path is*

$$\delta \to 3\delta \to 3\delta.$$

The game is solved by backward induction.

The dominant firm's final response. The dominant firm's final response solves

$$a_3^*(a_2) \in \arg\max_{a_3 \in \mathcal{P}} \pi_D(a_3, a_2).$$

The discount factor does not affect this final choice, because only the second market payoff remains. Given a rival price p, matching yields λp, while undercutting by one grid step yields $p - \delta$. Thus the dominant firm weakly prefers to match p if and only if

$$\lambda p \geq p - \delta,$$

or $p \leq \delta/(1-\lambda)$. Let

$$p^M \equiv \left\lfloor \frac{\delta}{1-\lambda} \right\rfloor_{\mathcal{P}}$$

denote the highest grid price the rival can induce the dominant firm to match.

The rival's response threshold. Suppose the dominant firm initially posts δ, and suppose $p^M > \delta$. The rival's response solves

$$a_2^*(a_1) \in \arg\max_{a_2 \in \mathcal{P}} \left\{ \pi_S(a_1, a_2) + \beta \pi_S\big(a_3^*(a_2), a_2\big) \right\}.$$

If the rival ties at δ, the lowest grid price, its payoff is $(1+\beta)(1-\lambda)\delta$. If the rival instead raises price to p^M, it loses demand at the first clearing but is matched by the dominant firm in the final revision, giving payoff $\beta(1-\lambda)p^M$. Lower price increases yield a smaller matched continuation price, while prices above p^M are not matched. The rival therefore prefers raising price to p^M whenever

$$\beta(1-\lambda)p^M > (1+\beta)(1-\lambda)\delta \quad \Longleftrightarrow \quad \beta > \bar{\beta}(\lambda) \equiv \frac{\delta}{p^M - \delta}.$$

Since $p^M = \lfloor \delta/(1-\lambda) \rfloor_{\mathcal{P}}$ is a step function that weakly increases in λ, the threshold $\bar{\beta}(\lambda)$ is weakly decreasing in λ.

Proof of Corollary 1. Suppose instead that the dominant firm's final turn is gated only after a rival price change. If the rival ties at δ, then $a_2 = a_1$, so the dominant firm's final turn is ordinary and ρ_D does not enter. The dominant firm has no reason to move away from the tie, so the rival's payoff from tying is $(1+\beta)(1-\lambda)\delta$. If the rival raises price to p^M, then $a_2 \neq a_1$, so the dominant firm's final turn is a response window. The rival is matched only when that window activates, giving expected payoff $\beta\rho_D(1-\lambda)p^M$. The rival prefers raising price to p^M whenever

$$\beta\rho_D(1-\lambda)p^M > (1+\beta)(1-\lambda)\delta \quad \Longleftrightarrow \quad \beta > \bar{\beta}(\lambda, \rho_D) \equiv \frac{\delta}{\rho_D\, p^M - \delta},$$

which reduces to $\bar{\beta}(\lambda)$ when $\rho_D = 1$. If instead $\rho_D p^M \leq \delta$, the left side is non-positive for any $\beta > 0$ while the right side is strictly positive, so the inequality fails for every β; the division above is invalid in this case, and no discount factor rationalizes initiation.

Verifying the equilibrium for a specific region. The threshold $\bar{\beta}(\lambda)$ is conditional on the dominant firm posting δ. It remains to verify that δ is the dominant firm's own best initial choice. Consider $\lambda \in (2/3, 3/4)$, so that $\delta/(1-\lambda)$ lies between 3δ and 4δ and $p^M = 3\delta$. Then $\bar{\beta} = 1/2$, so for $\beta > 1/2$ the rival's best response to an initial dominant price of δ is

$$a_2^*(\delta) = 3\delta.$$

The rival sacrifices current demand to create a higher matched continuation state.

Applying the same response problem to the other possible initial dominant prices, for $\beta > 1/2$ in this region, gives

$$a_2^*(2\delta) = \delta, \quad a_2^*(3\delta) = 2\delta, \quad a_2^*(4\delta) = 3\delta, \quad a_2^*(5\delta) = 4\delta.$$

Finally, the dominant firm's initial choice solves

$$a_1^* \in \arg\max_{a_1 \in \mathcal{P}} \left\{ \pi_D\big(a_1, a_2^*(a_1)\big) + \beta \pi_D\big(a_3^*(a_2^*(a_1)), a_2^*(a_1)\big) \right\}.$$

Given the response policy above, choosing δ gives the dominant firm current market demand and induces the rival to choose 3δ, which the dominant firm then matches. The resulting payoff, $\delta + 3\beta\lambda\delta$, strictly exceeds the payoff from every other initial price in this region. Hence, the unique equilibrium path is

$$\delta \to 3\delta \to 3\delta.$$

B.4 Reverse Timing in the Two-Clearing Game

Consider instead the order $S \to D \to M \to S \to M$, so that the rival receives the final revision opportunity. The rival's final response is the same follower problem behind p_S^{follow} in Proposition 1: it matches p if and only if $p \leq \delta/\lambda$. Since $\lambda > 1/2$, $\delta/\lambda < 2\delta$, so the rival's matching set never extends above the lowest grid price:

$$a_3^*(p) = \begin{cases} \delta, & p = \delta, \\ p - \delta, & p > \delta. \end{cases}$$

Consequently, whatever price the dominant firm is induced to match at the first clearing, the rival undoes it at the final revision whenever that price exceeds δ. For example, at $\lambda \in (2/3, 3/4)$ and $\beta < 3(1-\lambda)$, backward induction gives the equilibrium path

$$3\delta \to 3\delta \to 2\delta :$$

the rival initiates and the dominant firm matches at the first clearing, exactly as in the forward case, but the rival then undercuts its own initiated price at the final revision.

B.5 Proof of Proposition 3

Fix the candidate policy in Section 3.4. The proof verifies this policy by solving its continuation values and checking one-period deviations. Once the policy is fixed, each state has a value: current profit from the prescribed price choice plus the discounted value of the next state generated by the revision process. Thus, for firm i, the value of state (p_D, p_S, m, c) satisfies

$$V_i(p_D, p_S, m, c) = \pi_i\big(p'_D(\sigma_m), p'_S(\sigma_m)\big) + \beta\mathbb{E}\left[V_i(p'_D(\sigma_m), p'_S(\sigma_m), m', c')\right],$$

where only the active firm's price can change before the market clears, m' is the next scheduled mover, and c' records whether the active firm changed price. The expectation is over whether a response window activates. These Bellman equations are linear because the policy is fixed. I then check one-period deviations. Since there are only two prices, each deviation replaces the active firm's prescribed price with the other price and compares the resulting payoff with the value of following the policy.

Let $x = L/H$. Solving the Bellman system and checking one-period deviations gives four sufficient conditions. Two admit an exact closed form for every ρ_S:

$$x \leq \lambda, \qquad x \leq \beta.$$

The condition $x \leq \lambda$ is the dominant firm's matching condition: at (L, H), matching gives λH, while undercutting gives L. The condition $x \leq \beta$ is the rival's initiation condition from (L, L): the rival must care enough about the future matched price to give up current demand.

The remaining two conditions rule out, respectively, the dominant firm self-initiating H from (L, L) and the rival undercutting once the market reaches (H, H). Both depend on the continuation value of the off-path state (H, L), which loops back through the full state space rather than resolving in one step, so neither has a simple closed form for general ρ_S. At $\rho_S = 1$ (pure alternation) the loop shortens enough to give

$$\frac{\beta\lambda}{\lambda + \beta} \leq x \qquad \text{and} \qquad x \leq \frac{(1-\lambda)(1+\beta+\beta^2)}{1-\beta\lambda+\beta}.$$

For $\rho_S < 1$, I verify both conditions by solving the Bellman system directly rather than through a closed-form threshold. For example, at $(\lambda, \beta, \rho_D, \rho_S, x) = (0.7, 0.9, 1, 0, 0.65)$, all four deviation gaps are strictly positive. By continuity, the candidate policy is therefore optimal on a nonempty neighborhood of this point. Starting from (L, L), once the rival receives a revision opportunity it chooses H, the dominant firm then matches, and the market converges to (H, H).

The no-undercutting condition becomes easier to satisfy as ρ_S falls. A rival that undercuts must later pass through a response window to return to (H, H), so a less reliable rival expects a longer delay before it can restore the high price. This is why lower ρ_S weakly expands the set of parameters that sustain dominant followership.

B.6 Loyal-Demand Extension

The baseline model isolates the matching-versus-undercutting mechanism by assuming that a firm charging above its rival receives no demand. This extension relaxes that assumption by allowing each firm to retain loyal consumers even when it charges more than its rival. With loyal demand, the dominant follower can monetize demand advantage in two ways: it can match the rival's price, or it can remain above the rival and serve only loyal consumers. Matching must therefore beat not only undercutting, but also staying expensive.

Proposition 4 (Dominant followership with loyal demand)**.** *Fix a leader price $p = k\delta$, with $k \in \{2, \dots, M-1\}$. In the loyal-switcher environment, there exists a nonempty open set of primitives (α, β, λ), with*

$$0 < \beta < \alpha, \qquad \alpha + \beta < 1, \qquad \lambda \in [1/2, 1),$$

such that the dominant follower's unique best response is to match p, while the rival's unique best response is to undercut p by one grid step. Thus, the dominant-followership response pattern can arise even when both firms retain positive demand above the leader's price.

Figure A.3 shows that this can hold on a nonempty parameter region. Loyal demand narrows the mechanism, but does not eliminate it: when the dominant firm still gains enough from matching and the rival remains sufficiently dependent on switchers, the behavioral separation behind dominant followership survives.

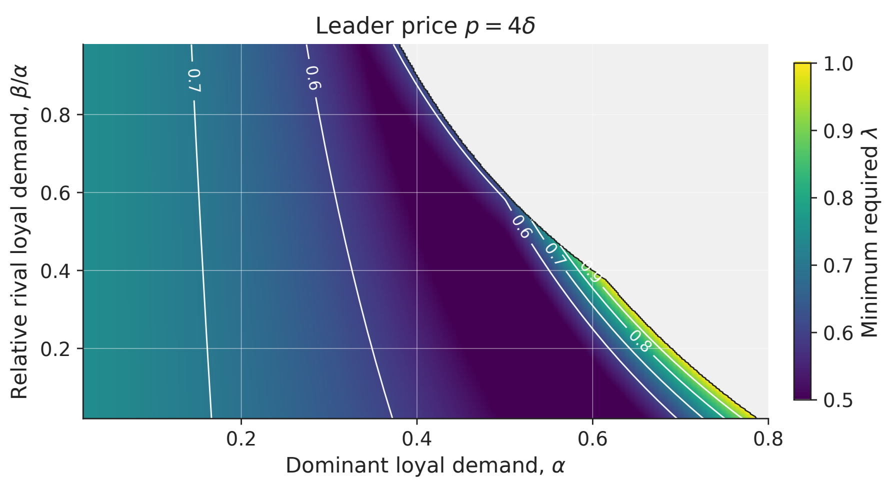

Figure A.3. Dominant Followership with Loyal Demand.

Notes: The figure fixes $M = 5$ and leader price $p = 4\delta$. It plots the minimum λ required for the dominant follower to match while the rival undercuts. Colored regions satisfy $\bar{\lambda}(\alpha, \beta, 4, 5) < 1$; gray regions admit no feasible $\lambda < 1$.

Proof of Proposition 4. Fix a leader price $p = k\delta$, with $k \in \{2, \dots, M-1\}$. Total demand is normalized to one. A mass α is loyal to the dominant firm, a mass β is loyal to the rival, and the remaining mass

$$s = 1 - \alpha - \beta$$

consists of switchers, with $0 < \beta < \alpha$ and $\alpha + \beta < 1$. Loyal consumers buy from their preferred firm regardless of relative prices. Switchers buy from the lower-price firm, and at equal prices the dominant firm receives share $\lambda \in [1/2, 1]$ of switchers. Demand for the dominant firm is therefore

$$q_D(p_D, p_S) = \begin{cases} \alpha + s, & p_D < p_S, \\ \alpha + \lambda s, & p_D = p_S, \\ \alpha, & p_D > p_S, \end{cases}$$

while demand for the rival is

$$q_S(p_D, p_S) = \begin{cases} \beta, & p_D < p_S, \\ \beta + (1-\lambda)s, & p_D = p_S, \\ \beta + s, & p_D > p_S. \end{cases}$$

The proof first reduces the follower's response problem to three relevant actions, then derives the response inequalities, and finally shows that the feasible set is nonempty.

Step 1: Relevant follower actions. For either follower, matching gives demand at price parity. Among prices below the leader, $p - \delta$ is optimal because any lower price receives the same demand at a lower margin. Among prices above the leader, $M\delta$ is optimal because the firm receives only loyal demand and profit is increasing in price. Thus, it is enough to compare three actions: match, undercut by one grid step, and choose the loyal-only high-price option.

For the dominant follower, the three payoffs are

$$\pi_D^{match} = k\delta[\alpha + \lambda s], \qquad \pi_D^{under} = (k-1)\delta(1-\beta), \qquad \pi_D^{high} = M\delta\alpha.$$

For the rival, they are

$$\pi_S^{match} = k\delta[\beta + (1-\lambda)s], \qquad \pi_S^{under} = (k-1)\delta(1-\alpha), \qquad \pi_S^{high} = M\delta\beta.$$

Step 2: Response inequalities. The dominant follower matches if matching beats both deviations:

$$\pi_D^{match} > \max\{\pi_D^{under}, \pi_D^{high}\}.$$

Substituting the payoffs gives

$$\lambda > \max\left\{\frac{(k-1)(1-\beta)-k\alpha}{ks}, \frac{\alpha(M-k)}{ks}\right\}.$$

The rival undercuts if undercutting beats both alternatives:

$$\pi_S^{under} > \max\{\pi_S^{match}, \pi_S^{high}\},$$

or equivalently,

$$\lambda > \frac{1-\alpha}{ks}$$

and

$$(k-1)(1-\alpha) > M\beta.$$

The first inequality makes undercutting better than matching; the second makes undercutting better than serving only loyal consumers at the high price.

Step 3: Feasibility. Combining the λ-restrictions gives the cutoff

$$\bar{\lambda}(\alpha,\beta,k,M) \equiv \max\left\{\frac{1}{2}, \frac{(k-1)(1-\beta)-k\alpha}{ks}, \frac{\alpha(M-k)}{ks}, \frac{1-\alpha}{ks}\right\},$$

where $s = 1-\alpha-\beta$. The dominant-followership response pattern holds whenever

$$\lambda > \bar{\lambda}(\alpha,\beta,k,M)$$

and

$$(k-1)(1-\alpha) > M\beta.$$

For fixed (α,β,k,M), an admissible $\lambda \in [1/2, 1)$ therefore exists if $\bar{\lambda}(\alpha,\beta,k,M) < 1$ and the additional condition holds.

These conditions are not vacuous. For any $\alpha \in (0, k/M)$, choosing $\beta > 0$ sufficiently small makes $\bar{\lambda}(\alpha, \beta, k, M) < 1$ and satisfies $(k-1)(1-\alpha) > M\beta$. Hence there exists $\lambda \in (\bar{\lambda}(\alpha, \beta, k, M), 1)$. Since the inequalities are strict, the response pattern holds on an open set of primitives.

C Simulation Appendix

C.1 Q-Learning Implementation Details

This appendix collects the exact specification of the Q-learning simulations summarized in Section 4.

Response opportunities. Suppose firm i is active in period t, so firm j is scheduled next. Let ρ_j denote the probability that firm j can revise on its next scheduled turn after firm i changes price. Draw $u_{t+1} \sim U[0,1]$ and define

$$R_{j,t+1} = \mathbf{1}\{a_{i,t} \neq p_{i,t-1}\}\mathbf{1}\{u_{t+1} \leq \rho_j\}.$$

If $a_{i,t} = p_{i,t-1}$, there is no rival price change, and period $t+1$ is firm j's ordinary pricing opportunity. If $a_{i,t} \neq p_{i,t-1}$ and $R_{j,t+1} = 1$, firm j's scheduled opportunity becomes a response, with $p_{j,t+1} = a_{j,t+1}$ and $p_{i,t+1} = p_{i,t}$. If $a_{i,t} \neq p_{i,t-1}$ and $R_{j,t+1} = 0$, no firm moves in period $t+1$, and the market clears again at unchanged posted prices. The ordinary alternating clock then continues; the main simulations vary ρ_D and ρ_S to distinguish symmetric from asymmetric response timing.

Exploration. Firms follow an ϵ_t-greedy policy,

$$a_{i,t} = \begin{cases} \arg\max_{a \in \mathcal{P}} Q_{i,t}(s_{i,t}, a), & \text{with probability } 1 - \epsilon_t, \\ \tilde{a}_{i,t}, \quad \tilde{a}_{i,t} \sim \text{Unif}(\mathcal{P}), & \text{with probability } \epsilon_t. \end{cases}$$

General learning target. Section 4 gives the Q-learning update rule and the simultaneous and two-turn posted-price targets. More generally, a scheduled response window may not be used, in which case the market still clears at unchanged posted prices and the firm earns another realized profit before its next pricing decision. Letting $m_i(t)$ denote the number of market clearings between firm i's choice at t and that next pricing decision, the implemented target is

$$\widehat{Q}_{i,t}^{post}(s_{i,t}, a_{i,t}) = \sum_{k=0}^{m_i(t)-1} \beta^k \pi_i(p_{i,t+k}, p_{j,t+k}) + \beta^{m_i(t)} \max_{a' \in \mathcal{P}} Q_{i,t}(s_{i,t+m_i(t)}, a').$$

This reduces to the two-turn target in Section 4 when $m_i(t) = 2$ and every scheduled window is used.

Parameters. The discount factor is fixed at $\beta = 0.99$. Learning rates and exploration rates follow parallel hyperbolic decay schedules over a simulation horizon of T periods:

$$\alpha_t = \max\left\{\underline{\alpha}, \alpha_0 \frac{\kappa_\alpha}{\kappa_\alpha + t}\right\}, \qquad \epsilon_t = \max\left\{\underline{\epsilon}, \epsilon_0 \frac{\kappa_\epsilon}{\kappa_\epsilon + t}\right\}.$$

In the baseline simulations, $\alpha_0 = \epsilon_0 = 0.1$, $\underline{\alpha} = \underline{\epsilon} = 0.05$, and $\kappa_\alpha = \kappa_\epsilon = 0.1T$, so both learning and exploration decline gradually over the simulation horizon but remain bounded away from zero.

Myopic best response. In the one-learner benchmarks, the myopic firm i chooses $BR_i(s_{i,t}) \in \arg\max_{p \in \mathcal{P}} \pi_i(p, p_{j,t-1})$: the price that maximizes current-period profit against the rival's last posted price, without updating continuation values.

C.2 No-Response Learning Benchmark

Figure A.4 plots the transaction-price trajectories for the simultaneous-updating benchmark discussed in the main text. With no response channel, demand asymmetry does not produce higher learned prices.

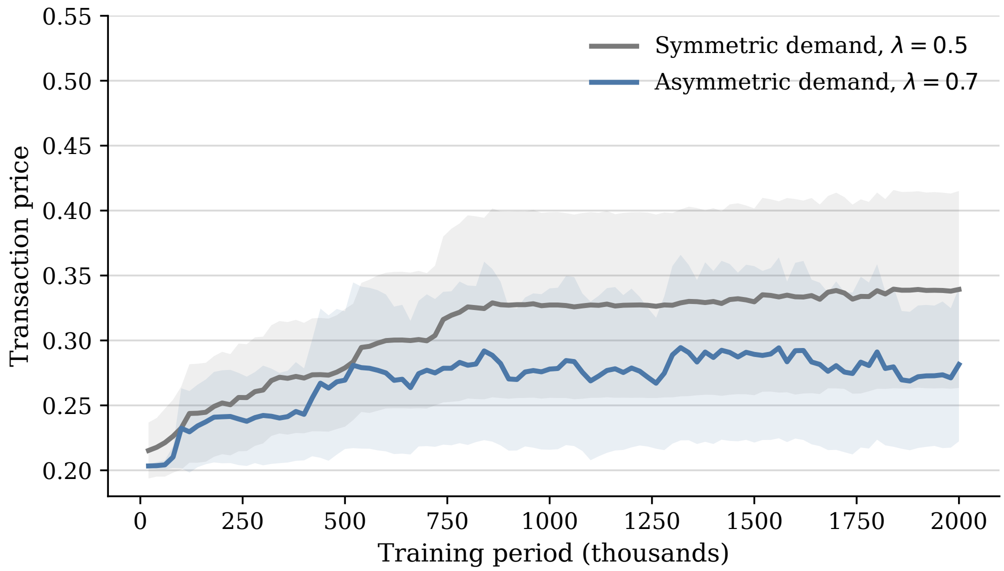

Figure A.4. Transaction Prices under Simultaneous Updating.

Notes: The figure plots average transaction prices in the simultaneous-updating benchmark. Lines average across simulation runs, and shaded regions report 95 percent confidence intervals.

C.3 Learning in the One-Clearing Sequential Benchmark

This appendix reports a one-clearing sequential learning benchmark. The benchmark mirrors the Stackelberg timing in Section 3.2, while removing the posted-price feature of the main learning simulations: prices do not generate demand before the follower responds. A leader chooses a price, the follower observes and responds, and the market clears once at the resulting price pair. The exercise therefore asks whether role assignment matters for learning even before adding the posted-price channel, in which prices remain active between revision opportunities.

Table A.4 reports long-run outcomes. With $\lambda = 0.7$, market prices are substantially higher when the dominant firm follows than when it leads. At the same time, the dominant-follower price remains close to the static one-clearing Stackelberg equilibrium price. For $\delta = 0.2$ and $\lambda = 0.7$, Proposition 1 gives $p_D^{follow} = 0.6$, close to the learned minimum market price of 0.619. This benchmark separates the role-asymmetry channel from the posted-price response channel used in the main simulations. In the stripped-down one-clearing environment, learning preserves the dominant-followership role asymmetry and remains close to the corresponding static benchmark. The main simulations then add the posted-price feature: prices remain active between revision opportunities and enter the states and rewards that algorithms learn from.

Table A.4. ONE-CLEARING SEQUENTIAL LEARNING BENCHMARK

Scenario	Leader price	Follower price	Minimum market price
Symmetric demand	0.661	0.662	0.636
Dominant follower $S \to D$	0.647	0.640	0.619
Dominant leader $D \to S$	0.430	0.454	0.378

Notes: Entries report average prices over the final 10 percent of learning periods across 30 simulation runs. The price grid is $\{0.2, 0.4, 0.6, 0.8, 1\}$. In asymmetric-demand rows, D denotes the dominant firm, S denotes the rival, and $\lambda = 0.7$.

The difference from the posted-price learning target in equation (3) is the payoff path generated by an initiated price. In the one-clearing benchmark, the leader's price generates no demand until after the follower responds. The inherited follower price can help define the state, but it is not an active price against which the leader earns current profit. The posted-price simulations add exactly this missing payoff: the initiated price first earns profit against the inherited rival price and then remains in the state to which the rival may respond.

C.4 Asymmetric Incidence across Runs

Figure A.5 plots the run-level counterpart to Table 5. The horizontal axis is the rival's posted-price premium, while the vertical axis is the dominant firm's profit premium. Points in the upper-right region therefore correspond to runs in which the rival posts higher prices on average and the dominant firm earns higher profits. The dominant-followership runs are concentrated in this region, showing that the asymmetric incidence in Table 5 is systematic across runs.

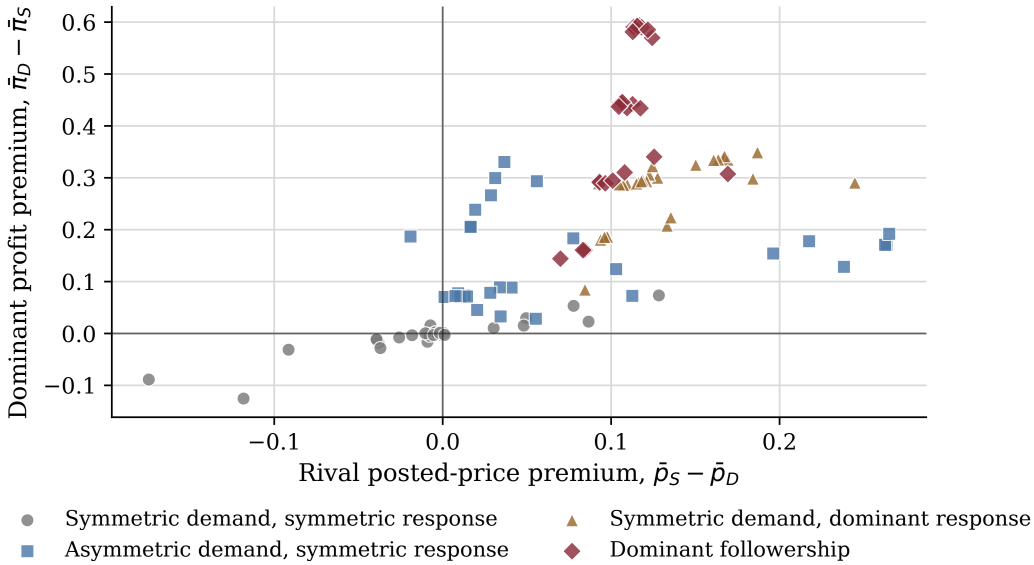

Figure A.5. Run-Level Incidence of Prices and Profits.

Notes: Each point is one simulation run, using averages over the final 5 percent of periods. The horizontal axis reports $\bar{p}_S - \bar{p}_D$, the difference between the rival's average posted price and the dominant firm's average posted price. The vertical axis reports $\bar{\pi}_D - \bar{\pi}_S$, the corresponding dominant-firm profit premium.

C.5 Response Decomposition

Figure A.6 decomposes the response-side firm's next-period action after the other firm posts above its inherited price. The main text focuses on matching because matching is the response that reinforces the initiator's higher-price move. This appendix figure shows what happens when matching does not occur. In the comparison cells, higher-price moves are often followed by undercutting or no response. When demand advantage and asymmetric response timing are combined, matching accounts for most of the response-side firm's next-period actions and undercutting becomes much less frequent.

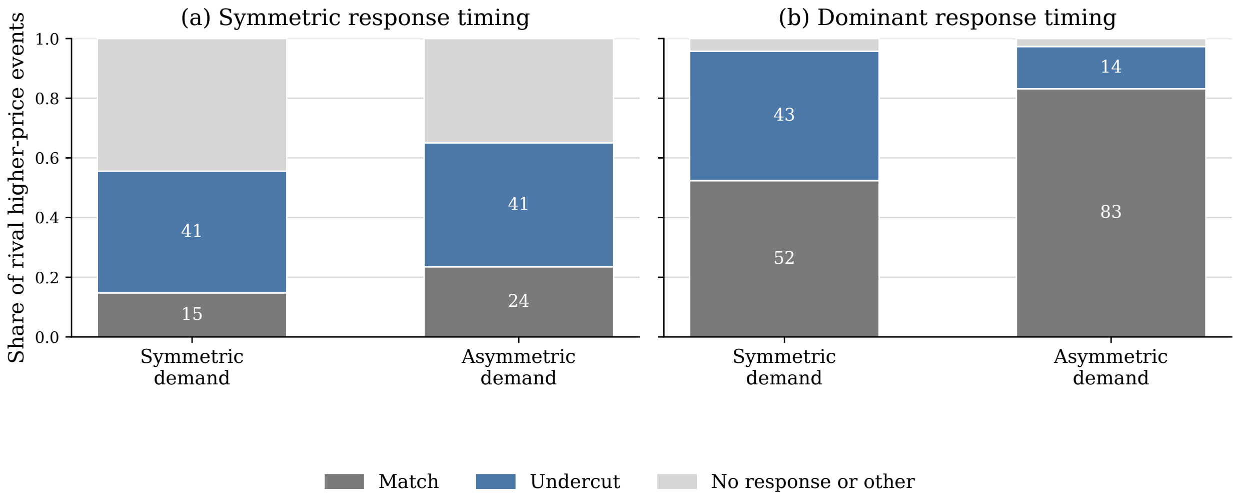

Figure A.6. Response Decomposition after Rival Higher Prices.

Notes: The figure conditions on final-5-percent periods in which one firm posts a price above the response-side firm's inherited posted price. Bars classify the response-side firm's next-period action. A match means the response-side firm sets the same price as the initiator; an undercut means it sets a lower price. The remaining category includes cases in which the response-side firm's next posted price is unchanged or is above the initiator's price. In asymmetric-demand cells, the response-side firm is the demand-advantaged firm.

C.6 Learning into High-Price States

Figure A.7 reports the share of periods in which both firms post prices weakly above 0.6, by training bin. The figure complements the final-period state distribution in Figure 8 by showing how the share of high-price states evolves over the course of training. Asymmetric response timing is associated with earlier movement into high-price states, and the share is highest when the reliable responder is also demand advantaged.

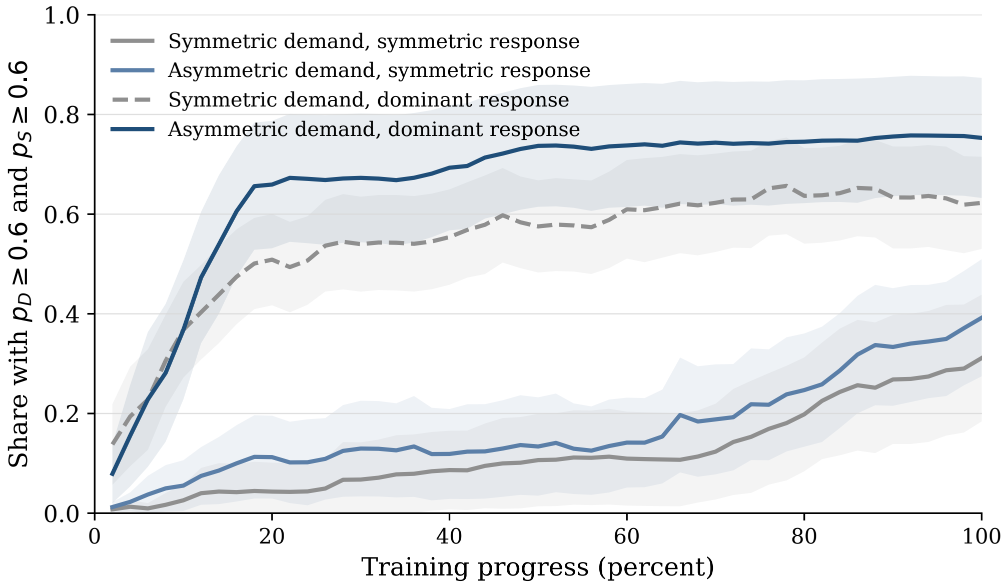

Figure A.7. Learning into High-Price States.

Notes: The figure reports, by training bin, the share of periods in which both firms post prices weakly above 0.6. Lines average across simulation runs; shaded regions report 95 percent confidence intervals across runs.